

Let's Practice Subnetting

192.168.7.0 / 24

Need 2 subnets

Start - 192.168.7.0 / 24

Need 2 subnets

What is the new subnet mask?

What are the subnet IDs?

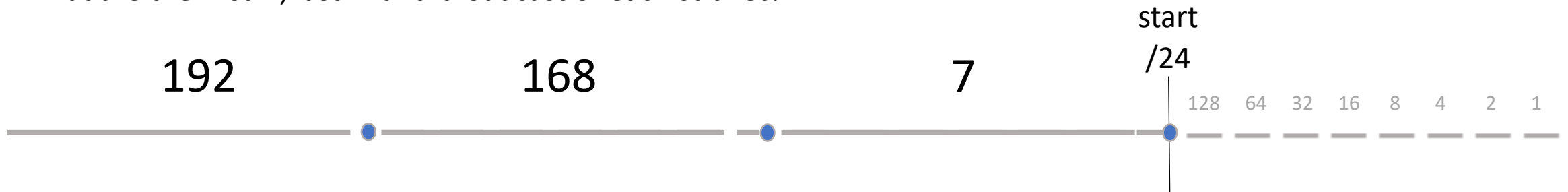
With the new mask, how many hosts per subnet?

What are the first IP, last IP and broadcast of each subnet?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



Start - 192.168.7.0 / 24

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<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

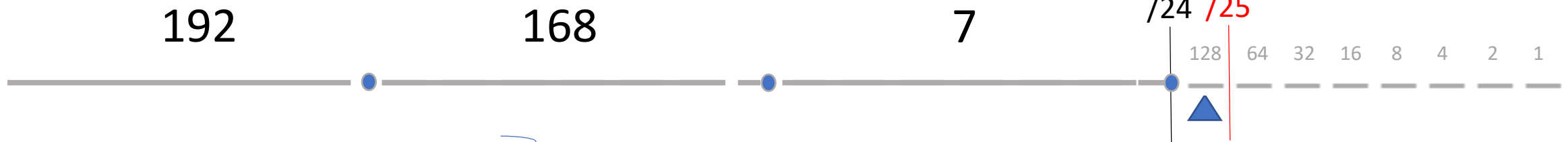
How many you need

Bits to move the mask

255.255.255.128

start new

/24 /25



192.168.7.0

192.168.7.256

Next number
Out of bounds

192.168.7.1

192.168.7.126

192.168.7.127

192.168.7.128

192.168.7.129

192.168.7.254

192.168.7.255

Start - 192.168.7.0 / 24

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7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

255.255.255.128

start new

/24 /25

128 64 32 16 8 4 2 1

192

168

7

192.168.7.0

192.168.7.1

192.168.7.126

192.168.7.127

192.168.7.128

192.168.7.129

192.168.7.254

192.168.7.255

192.168.29.0 / 24

Need 8 subnets

Start - 192.168.29.0 / 24

Need 8 subnets

What is the new subnet mask?

What are the subnet IDs?

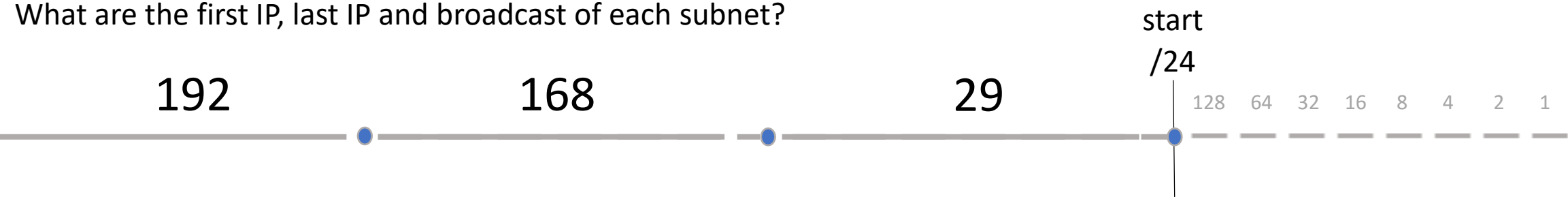
With the new mask, how many hosts per subnet?

What are the first IP, last IP and broadcast of each subnet?

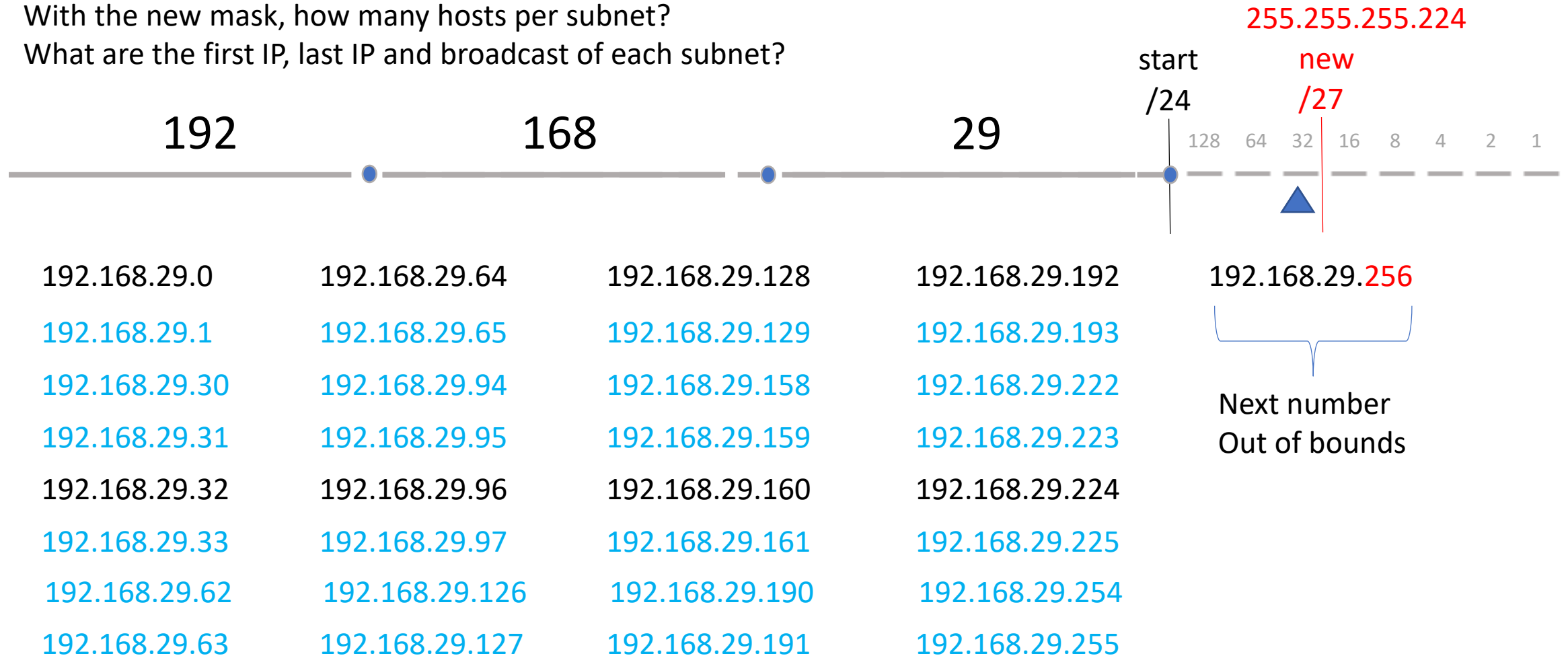
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



What are the first IP, last IP and broadcast of each subnet?



Start - 192.168.29.0 / 24

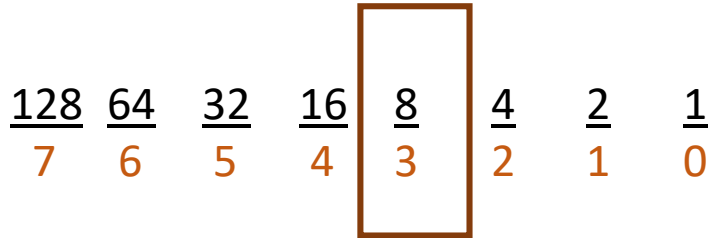
Need 8 subnets

What is the new subnet mask?

What are the subnet IDs?

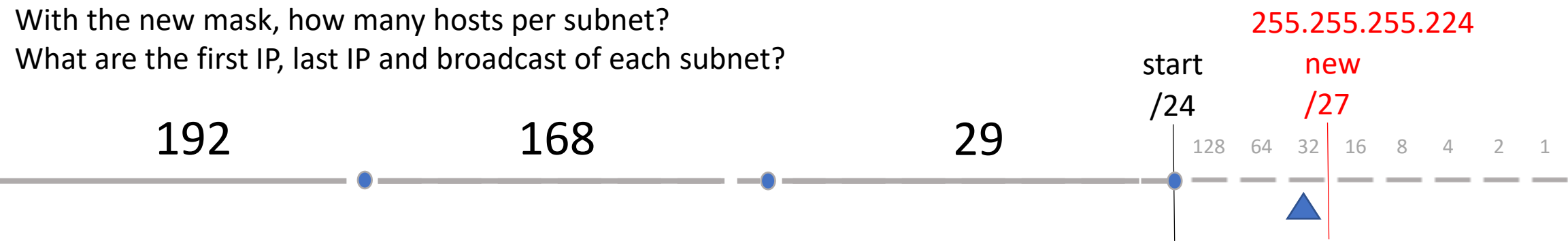
With the new mask, how many hosts per subnet?

What are the first IP, last IP and broadcast of each subnet?



How many you need

Bits to move the mask



192.168.29.0	192.168.29.64	192.168.29.128	192.168.29.192
192.168.29.1	192.168.29.65	192.168.29.129	192.168.29.193
192.168.29.30	192.168.29.94	192.168.29.158	192.168.29.222
192.168.29.31	192.168.29.95	192.168.29.159	192.168.29.223
192.168.29.32	192.168.29.96	192.168.29.160	192.168.29.224
192.168.29.33	192.168.29.97	192.168.29.161	192.168.29.225
192.168.29.62	192.168.29.126	192.168.29.190	192.168.29.254
192.168.29.63	192.168.29.127	192.168.29.191	192.168.29.255

192.168.200.0 / 24

Need 4 subnets

Start - 192.168.200.0 / 24

Need 4 subnets

What is the new subnet mask?

What are the subnet IDs?

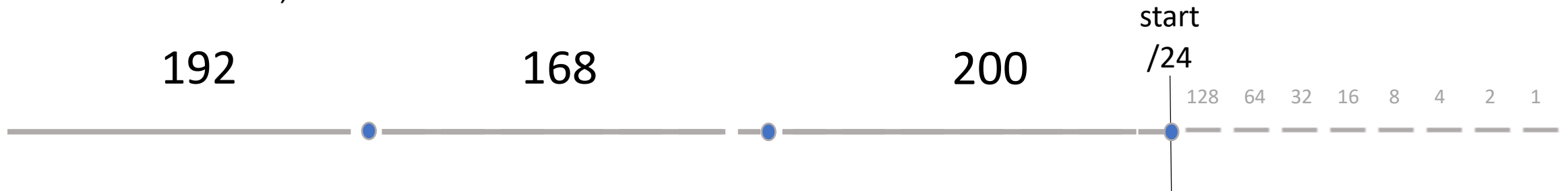
With the new mask, how many hosts per subnet?

What are the first IP, last IP and broadcast of each subnet?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



Start - 192.168.200.0 / 24

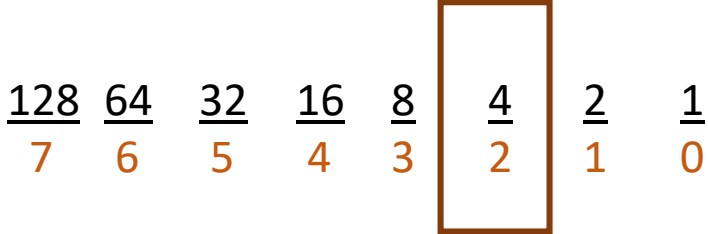
Need 4 subnets

What is the new subnet mask?

What are the subnet IDs?

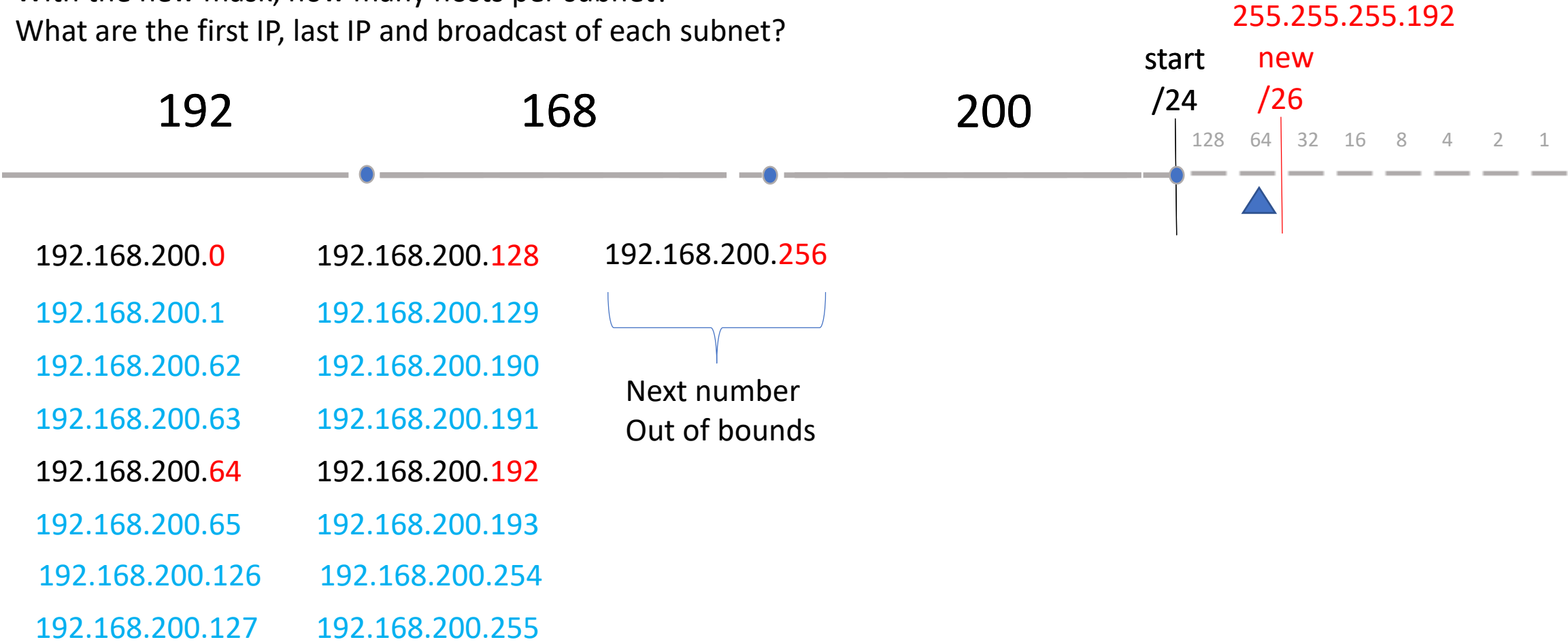
With the new mask, how many hosts per subnet?

What are the first IP, last IP and broadcast of each subnet?



How many you need

Bits to move the mask



Start - 192.168.200.0 / 24

Need 4 subnets

What is the new subnet mask?

What are the subnet IDs?

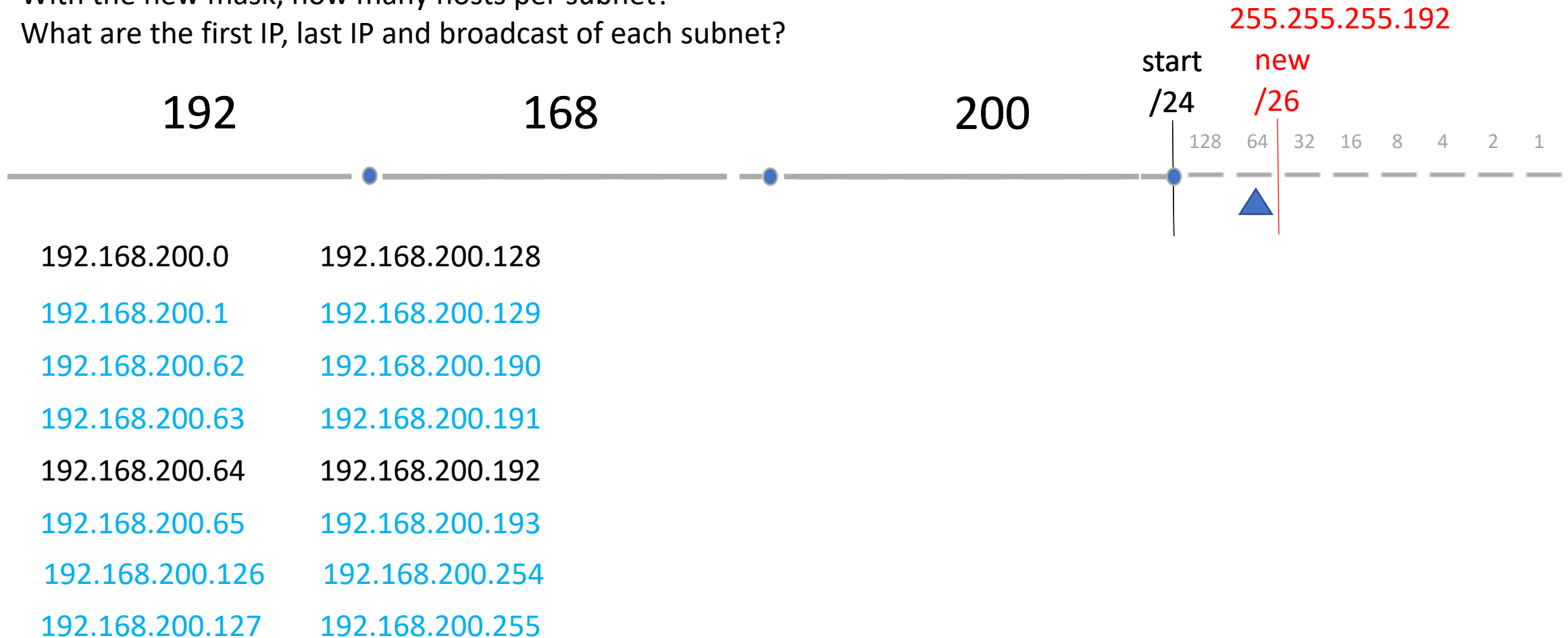
With the new mask, how many hosts per subnet?

What are the first IP, last IP and broadcast of each subnet?

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



192.168.44.0 / 24

Need as many point-to-point WAN links
as possible

Start - 192.168.44.0 / 24

Need as many point-to-point WAN links as possible

What is the new subnet mask?

How many hosts can each WAN link have?

How many WAN links can you have?

What are the subnet IDs for the first 4 subnets?

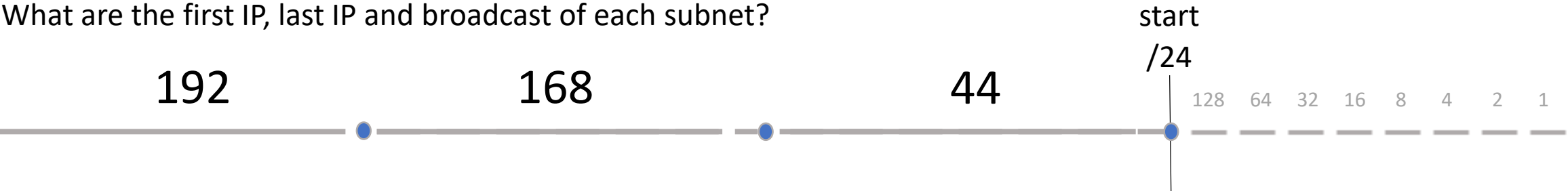
What are the subnet IDs for the last 4 subnets?

What are the first IP, last IP and broadcast of each subnet?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



Start - 192.168.44.0 / 24

Need as many point-to-point WAN links as possible

What is the new subnet mask?

How many hosts can each WAN link have?

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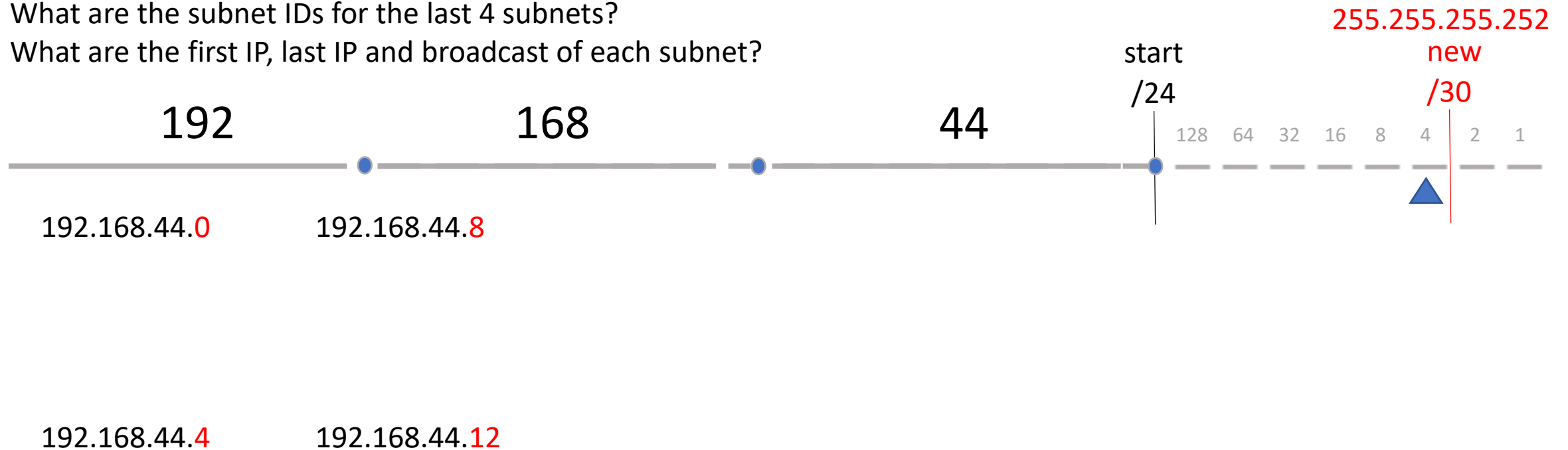
What are the subnet IDs for the last 4 subnets?

What are the first IP, last IP and broadcast of each subnet?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



Start - 192.168.44.0 / 24

Need as many point-to-point WAN links as possible

What is the new subnet mask?

How many hosts can each WAN link have?

How many WAN links can you have?

What are the subnet IDs for the first 4 subnets?

What are the subnet IDs for the last 4 subnets?

What are the first IP, last IP and broadcast of each subnet?

128

7

64

32

16

8

4

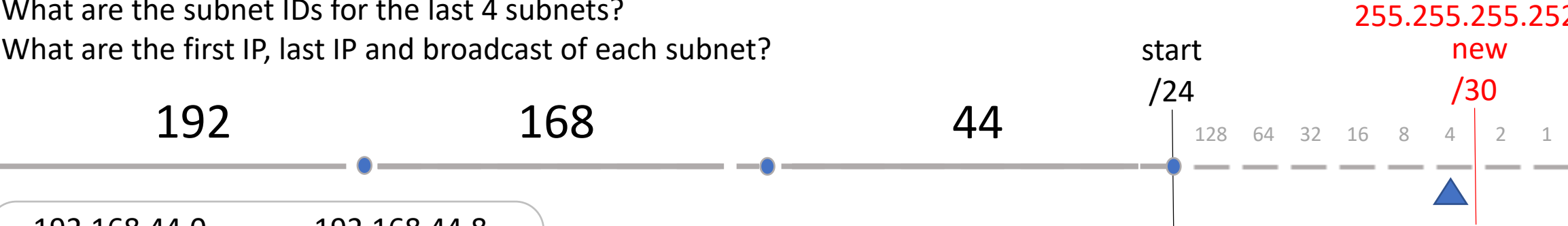
2

2

1

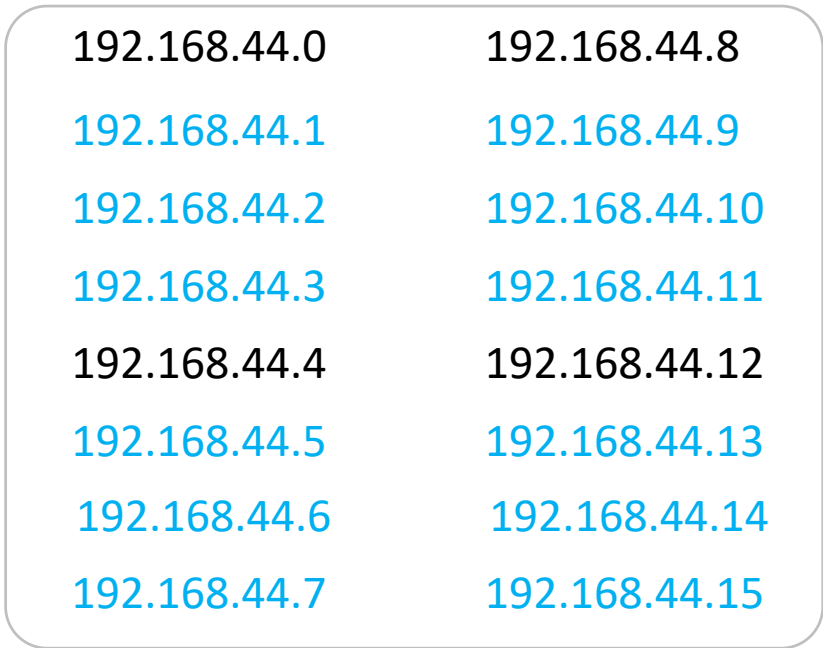
How many you need

Bits to move the mask



192.168.44.0	192.168.44.8
192.168.44.1	192.168.44.9
192.168.44.2	192.168.44.10
192.168.44.3	192.168.44.11
192.168.44.4	192.168.44.12
192.168.44.5	192.168.44.13
192.168.44.6	192.168.44.14
192.168.44.7	192.168.44.15

What are the first IP, last IP and broadcast of each subnet?



Start - 192.168.44.0 / 24

Need as many point-to-point WAN links as possible

What is the new subnet mask?

How many hosts can each WAN link have?

How many WAN links can you have?

What are the subnet IDs for the first 4 subnets?

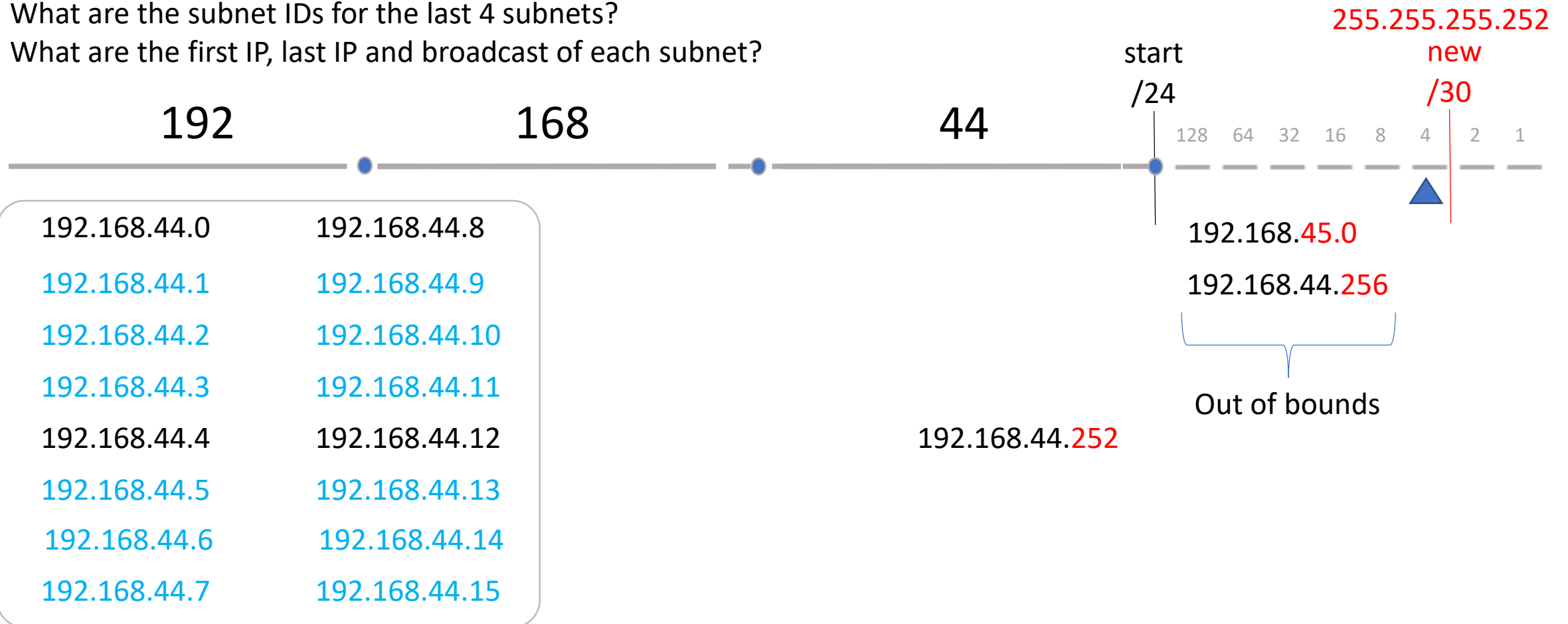
What are the subnet IDs for the last 4 subnets?

What are the first IP, last IP and broadcast of each subnet?

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



Start - 192.168.44.0 / 24

Need as many point-to-point WAN links as possible

What is the new subnet mask?

How many hosts can each WAN link have?

How many WAN links can you have?

What are the subnet IDs for the first 4 subnets?

What are the subnet IDs for the last 4 subnets?

What are the first IP, last IP and broadcast of each subnet?

128

7

64

6

32

5

16

4

8

3

4

2

2

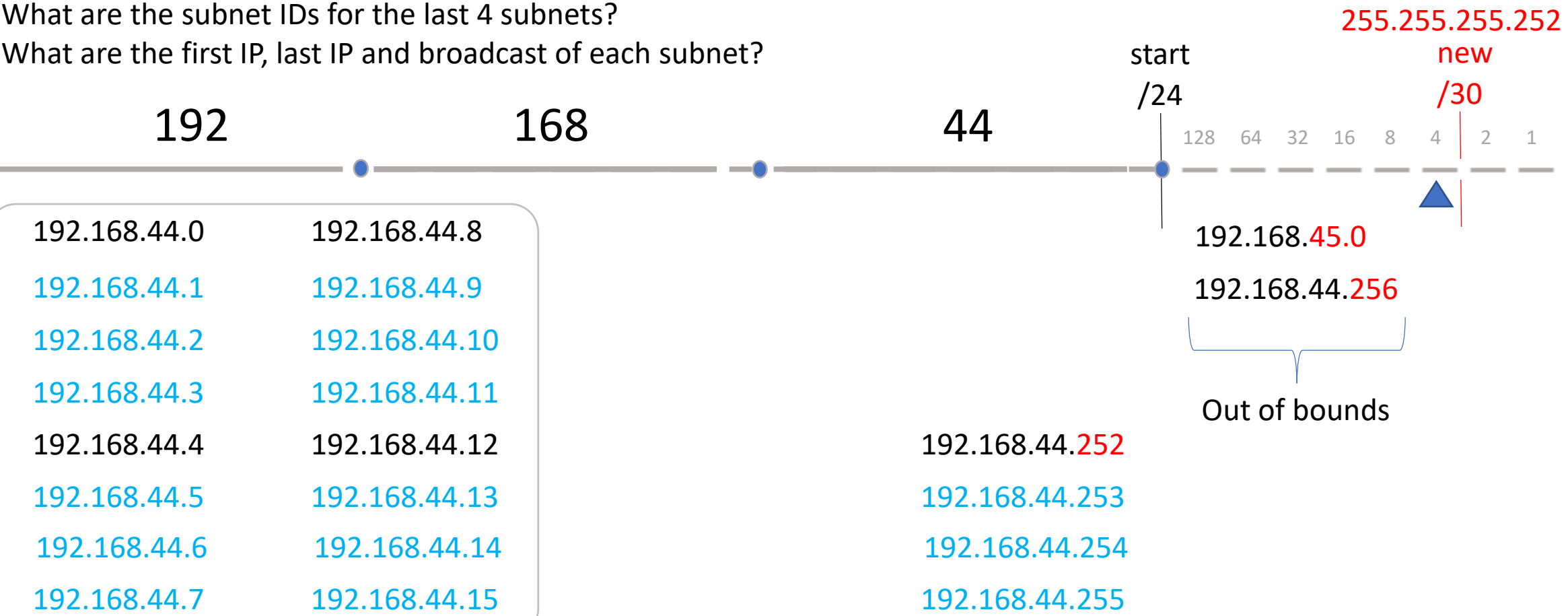
1

1

0

How many you need

Bits to move the mask



Start - 192.168.44.0 / 24

Need as many point-to-point WAN links as possible

What is the new subnet mask?

How many hosts can each WAN link have?

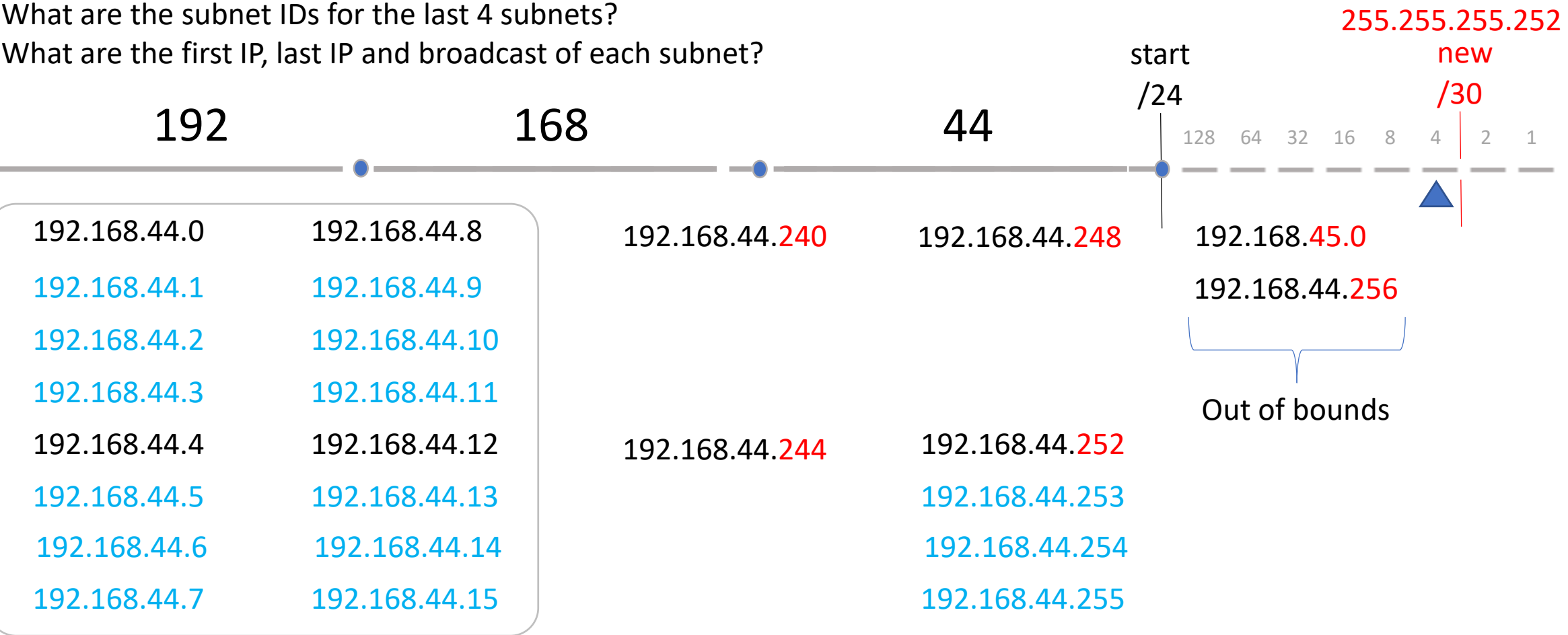
How many WAN links can you have?

What are the subnet IDs for the first 4 subnets?

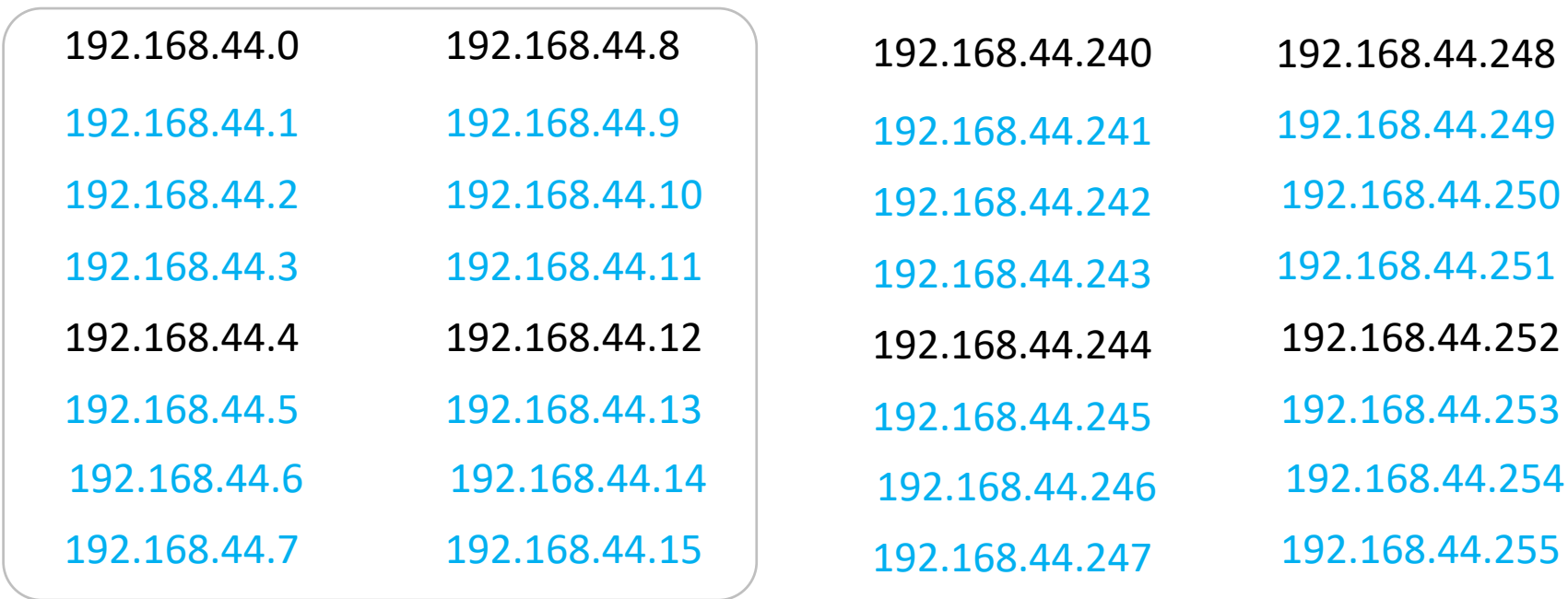
What are the subnet IDs for the last 4 subnets?

What are the first IP, last IP and broadcast of each subnet?

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$	How many you need
								Bits to move the mask



What are the first IP, last IP and broadcast of each subnet?



172.16.0.0 /16

Need 16 subnets

Start – 172.16.0.0 /16

What are the first and last IPs in this network?

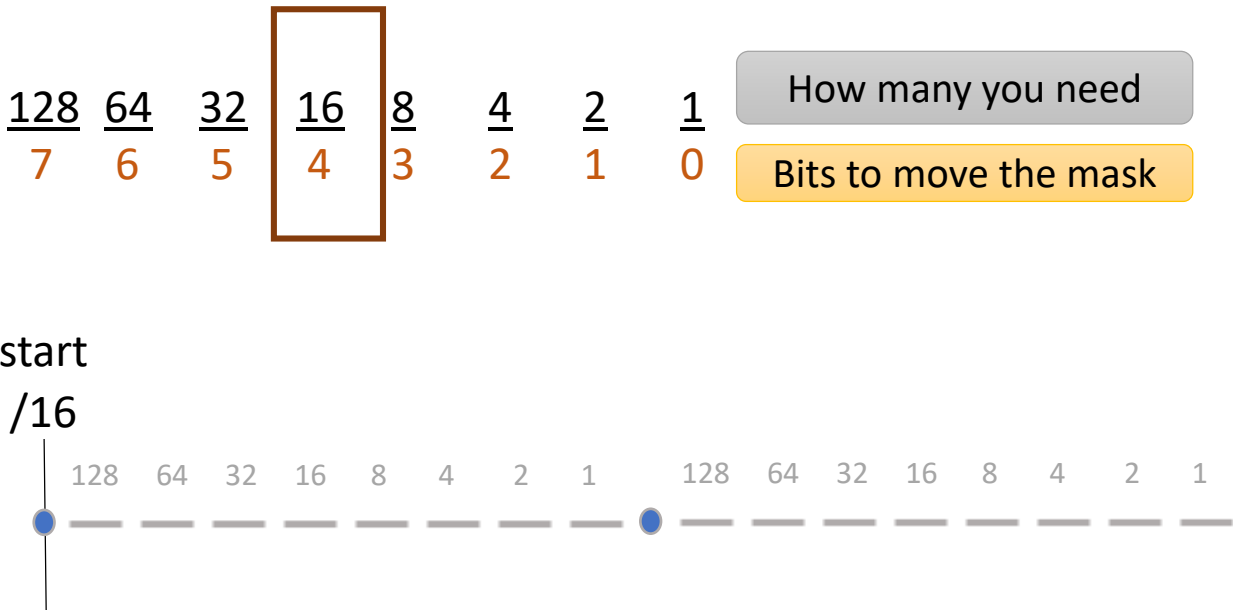
Need 16 subnets

What is the new subnet mask?

With the new mask, how many hosts per subnet?

What are the subnet IDs for the first 4 subnets?

What are the subnet IDs for the last 4 subnets?



Start – 172.16.0.0 /16

What are the first and last IPs in this network?

Need 16 subnets

What is the new subnet mask?

With the new mask, how many hosts per subnet?

What are the subnet IDs for the first 4 subnets?

What are the subnet IDs for the last 4 subnets?

128

64

32

16

8

4

2

1

7

6

5

4

3

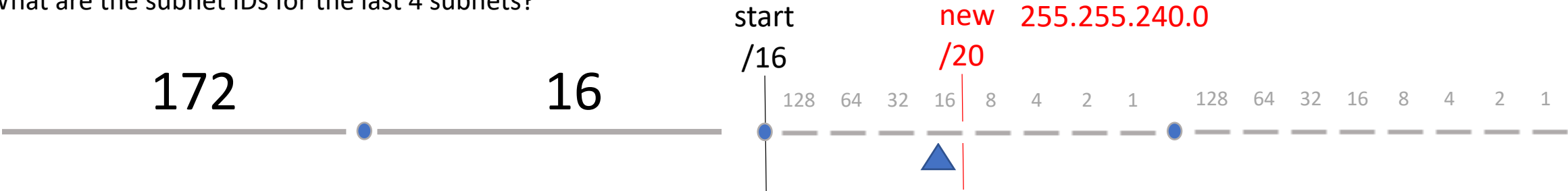
2

1

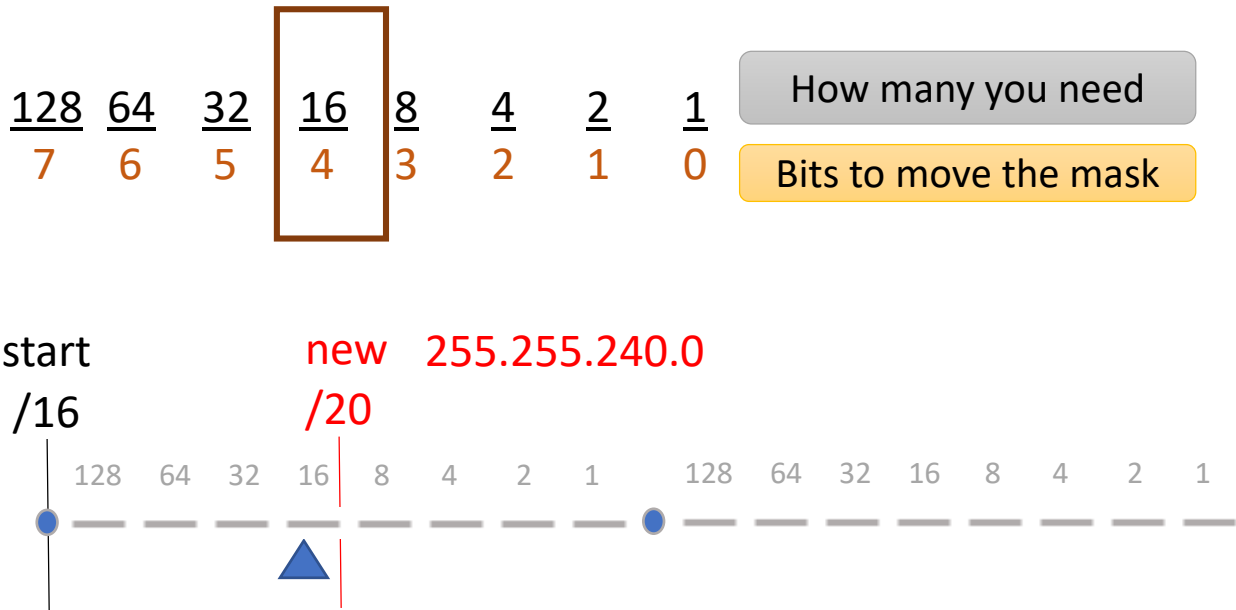
0

How many you need

Bits to move the mask

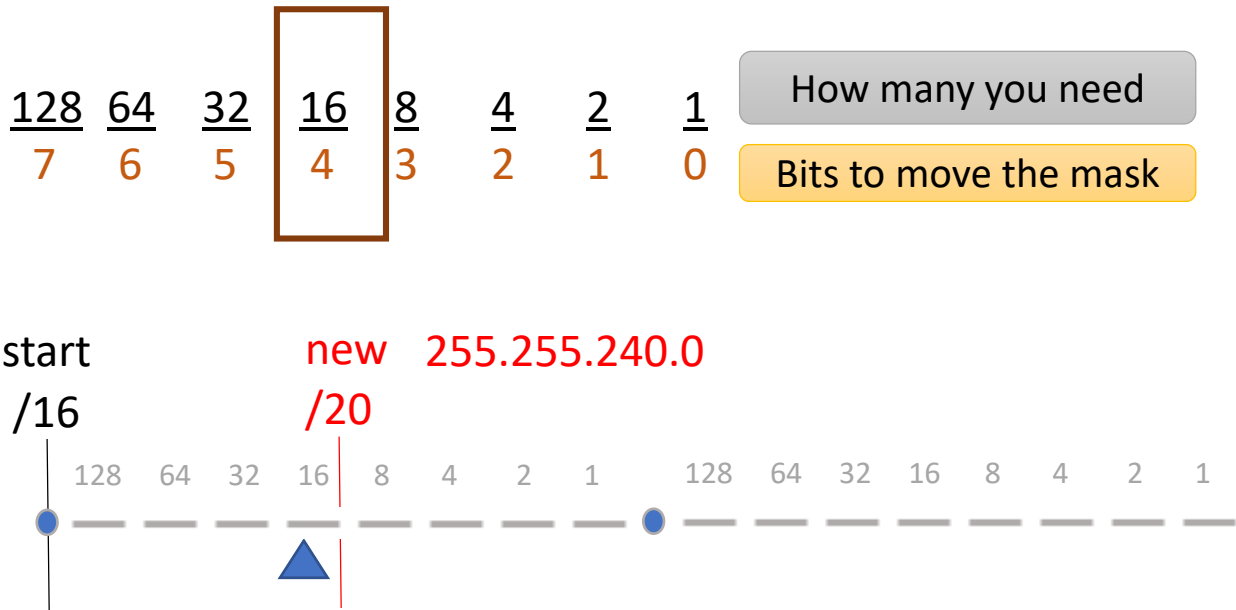


Start – 172.16.0.0 /16
What are the first and last IPs in this network?
Need 16 subnets
What is the new subnet mask?
With the new mask, how many hosts per subnet?
What are the subnet IDs for the first 4 subnets?
What are the subnet IDs for the last 4 subnets?



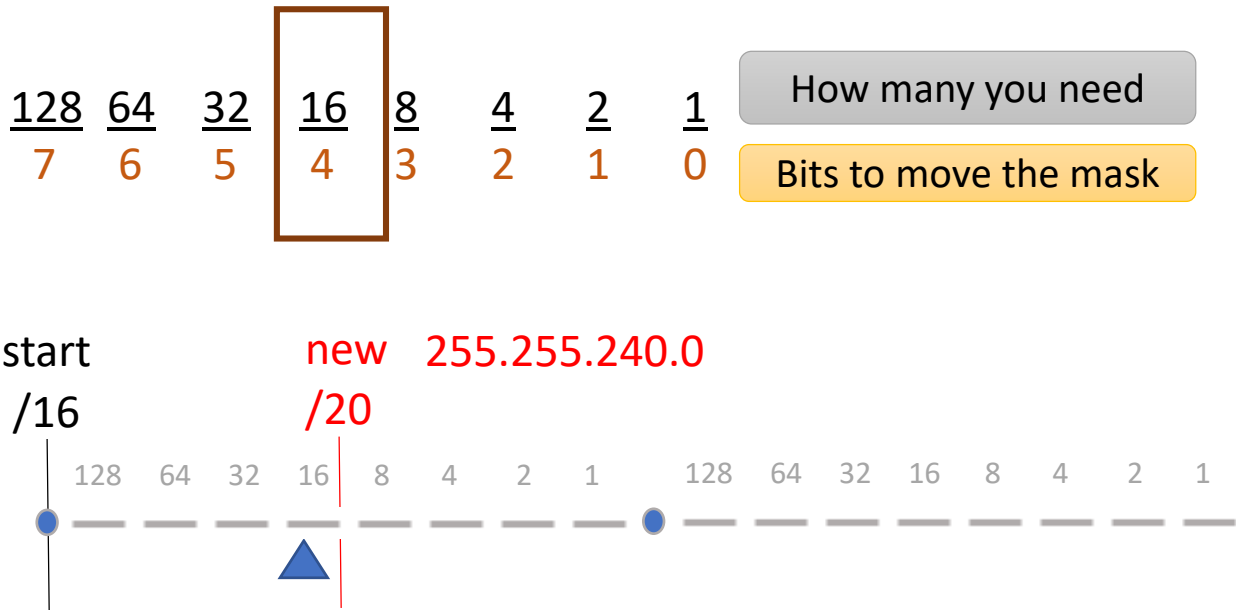
- | | |
|---------------|---------------|
| 172.16.0.0 | 172.16.32.0 |
| 172.16.0.1 | 172.16.32.1 |
| 172.16.15.254 | 172.16.47.254 |
| 172.16.15.255 | 172.16.47.255 |
| 172.16.16.0 | 172.16.48.0 |
| 172.16.16.1 | 172.16.48.1 |
| 172.16.31.254 | 172.16.63.254 |
| 172.16.31.255 | 172.16.63.255 |
| | 172.16.64.0 |

Start – 172.16.0.0 /16
What are the first and last IPs in this network?
Need 16 subnets
What is the new subnet mask?
With the new mask, how many hosts per subnet?
What are the subnet IDs for the first 4 subnets?
What are the subnet IDs for the last 4 subnets?



172.16.0.0	172.16.32.0
172.16.0.1	172.16.32.1
172.16.15.254	172.16.47.254
172.16.15.255	172.16.47.255
172.16.16.0	172.16.48.0
172.16.16.1	172.16.48.1
172.16.31.254	172.16.63.254
172.16.31.255	172.16.63.255

Start – 172.16.0.0 /16
What are the first and last IPs in this network?
Need 16 subnets
What is the new subnet mask?
With the new mask, how many hosts per subnet?
What are the subnet IDs for the first 4 subnets?
What are the subnet IDs for the last 4 subnets?

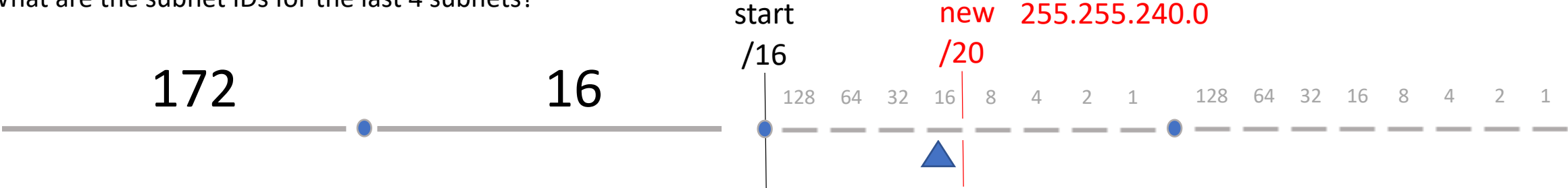
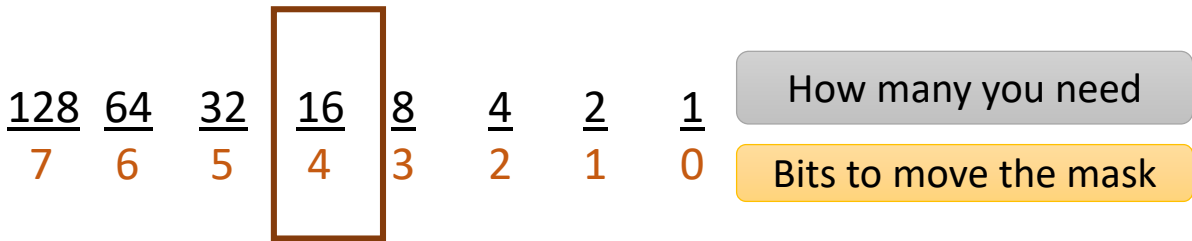


172.16.0.0	172.16.32.0
172.16.0.1	172.16.32.1
172.16.15.254	172.16.47.254
172.16.15.255	172.16.47.255
172.16.16.0	172.16.48.0
172.16.16.1	172.16.48.1
172.16.31.254	172.16.63.254
172.16.31.255	172.16.63.255

172.17.0.0
172.16.256.0
Out of bounds

172.16.240.0
172.16.240.1
172.16.255.254
172.16.255.255

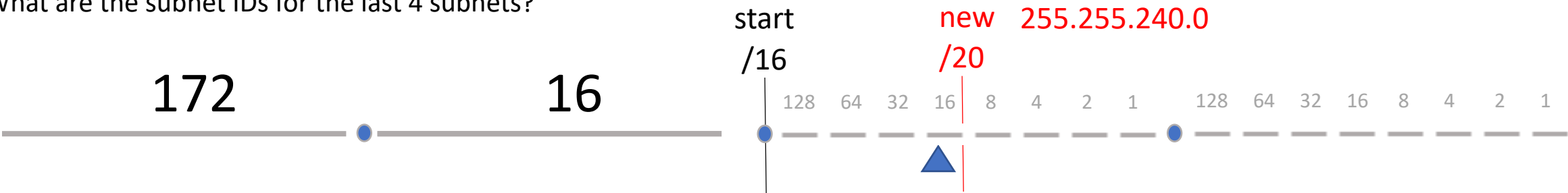
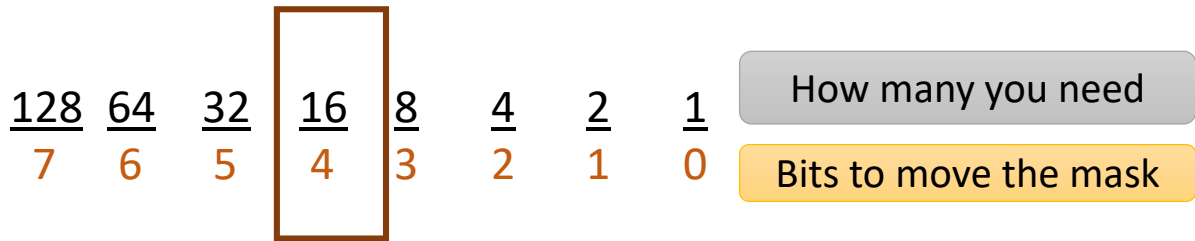
Start – 172.16.0.0 /16
What are the first and last IPs in this network?
Need 16 subnets
What is the new subnet mask?
With the new mask, how many hosts per subnet?
What are the subnet IDs for the first 4 subnets?
What are the subnet IDs for the last 4 subnets?



172.16.0.0	172.16.32.0
172.16.0.1	172.16.32.1
172.16.15.254	172.16.47.254
172.16.15.255	172.16.47.255
172.16.16.0	172.16.48.0
172.16.16.1	172.16.48.1
172.16.31.254	172.16.63.254
172.16.31.255	172.16.63.255

172.16.192.0	172.16.224.0
172.16.208.0	172.16.240.0
	172.16.240.1
	172.16.255.254
	172.16.255.255

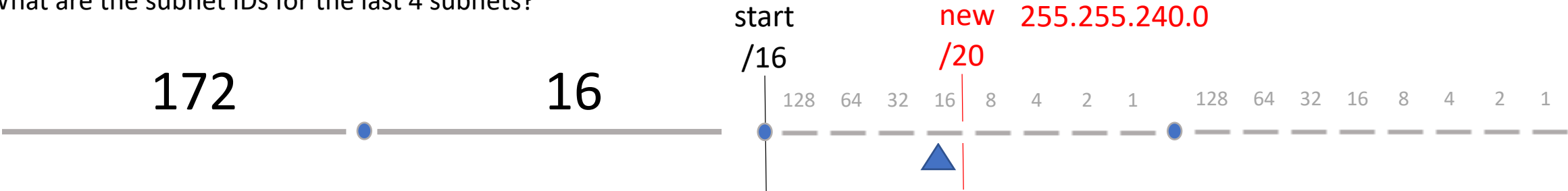
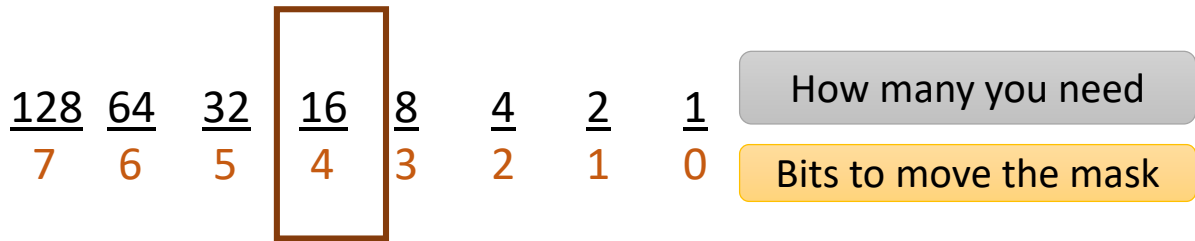
Start – 172.16.0.0 /16
What are the first and last IPs in this network?
Need 16 subnets
What is the new subnet mask?
With the new mask, how many hosts per subnet?
What are the subnet IDs for the first 4 subnets?
What are the subnet IDs for the last 4 subnets?



172.16.0.0	172.16.32.0
172.16.0.1	172.16.32.1
172.16.15.254	172.16.47.254
172.16.15.255	172.16.47.255
172.16.16.0	172.16.48.0
172.16.16.1	172.16.48.1
172.16.31.254	172.16.63.254
172.16.31.255	172.16.63.255

172.16.192.0	172.16.224.0
172.16.192.1	172.16.224.1
172.16.207.254	172.16.239.254
172.16.207.255	172.16.239.255
172.16.208.0	172.16.240.0
172.16.208.1	172.16.240.1
172.16.223.254	172.16.255.254
172.16.223.255	172.16.255.255

Start – 172.16.0.0 /16
What are the first and last IPs in this network?
Need 16 subnets
What is the new subnet mask?
With the new mask, how many hosts per subnet?
What are the subnet IDs for the first 4 subnets?
What are the subnet IDs for the last 4 subnets?



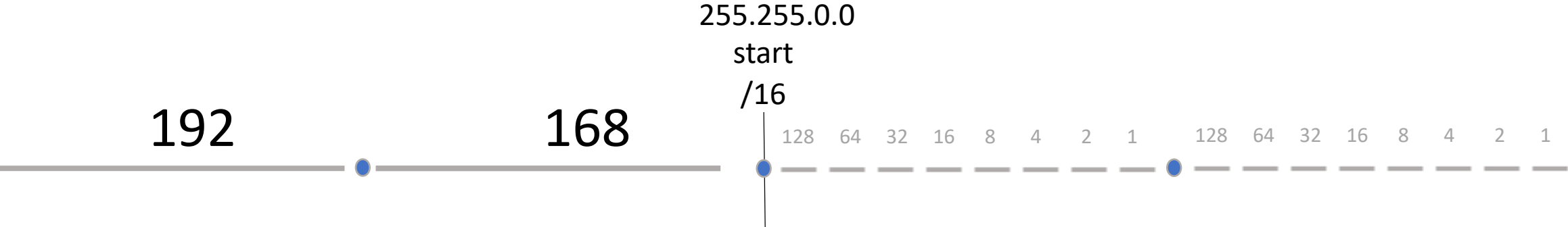
172.16.0.0	172.16.32.0	172.16.192.0	172.16.224.0
172.16.0.1	172.16.32.1	172.16.192.1	172.16.224.1
172.16.15.254	172.16.47.254	172.16.207.254	172.16.243.254
172.16.15.255	172.16.47.255	172.16.207.255	172.16.243.255
172.16.16.0	172.16.48.0	172.16.208.0	172.16.240.0
172.16.16.1	172.16.48.1	172.16.208.1	172.16.240.1
172.16.31.254	172.16.63.254	172.16.223.254	172.16.255.254
172.16.31.255	172.16.63.255	172.16.223.255	172.16.255.255

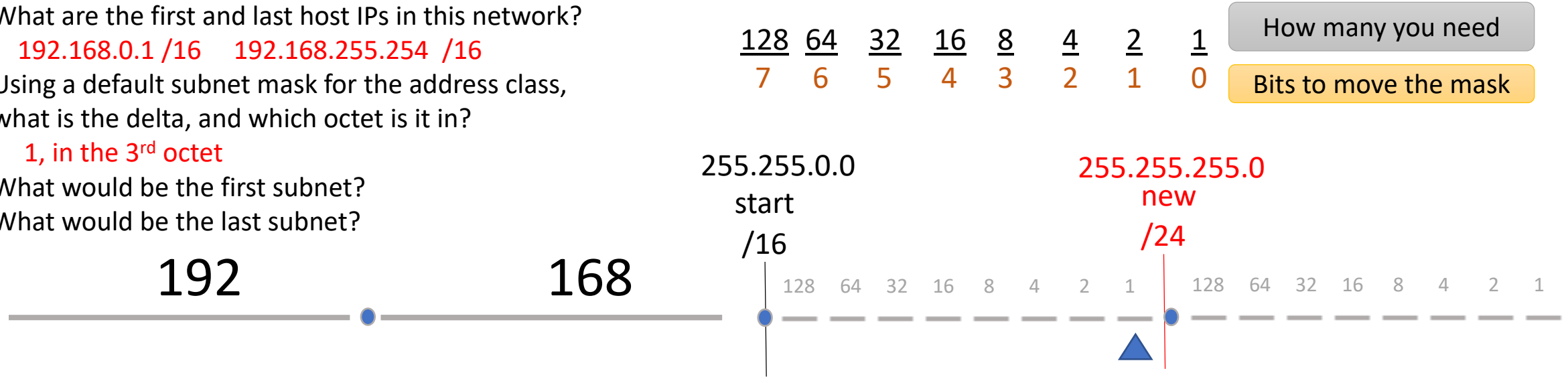
Examining Private IP Address Block 192.168.0.0 /16

Start – 192.168.0.0 /16
What are the first and last host IPs in this network?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	How many you need
7	6	5	4	3	2	1	0	

Bits to move the mask





Start – 192.168.0.0 /16
What are the first and last host IPs in this network?
192.168.0.1 /16 192.168.255.254 /16

Using a default subnet mask for the address class,
what is the delta, and which octet is it in?

1, in the 3rd octet
Using the new mask, what is the first subnet?
What is the last subnet?

128
7

64
6

32
5

16
4

8
3

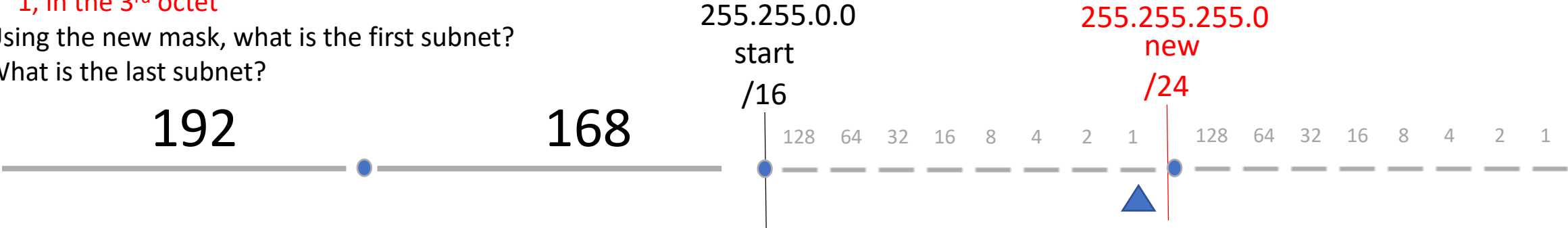
4
2

2
1

1
0

How many you need

Bits to move the mask



- 192.168.0.0

192.168.255.0
- 192.168.0.1

192.168.255.1
- 192.168.0.254

192.168.255.254
- 192.168.0.255

192.168.255.255
- 192.169.0.0

Out of bounds

Start – 192.168.0.0 /16

What are the first and last host IPs in this network?

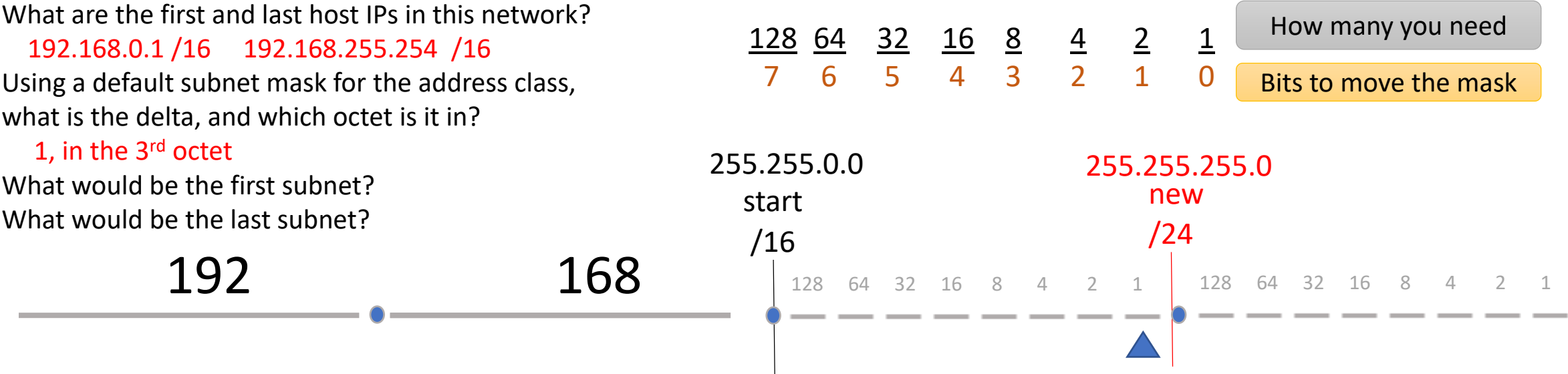
192.168.0.1 /16 192.168.255.254 /16

Using a default subnet mask for the address class, what is the delta, and which octet is it in?

1, in the 3rd octet

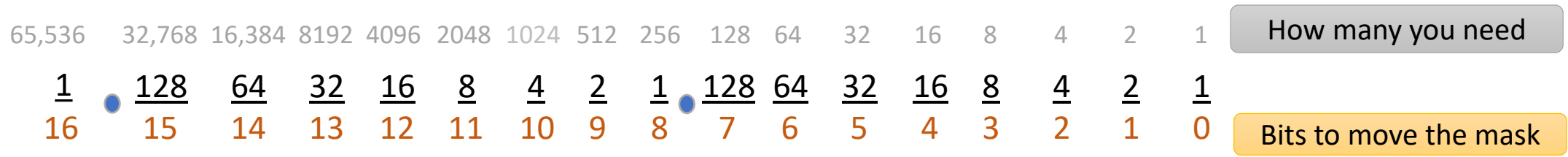
What would be the first subnet?

What would be the last subnet?

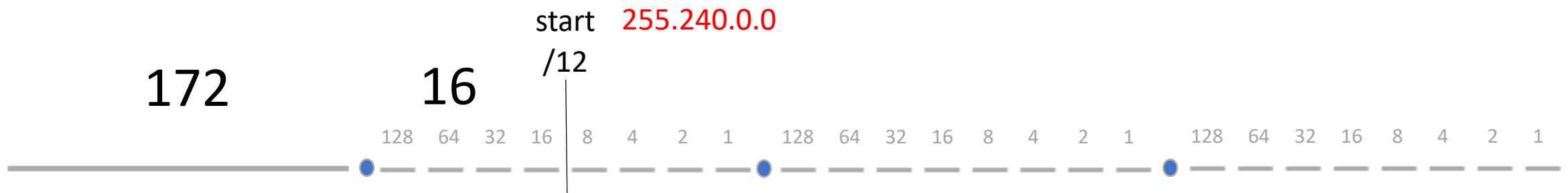


- | | |
|---------------|-----------------|
| 192.168.0.0 | 192.168.255.0 |
| 192.168.0.1 | 192.168.255.1 |
| 192.168.0.254 | 192.168.255.254 |
| 192.168.0.255 | 192.168.255.255 |

Examining Private IP Address Block 172.16.0.0 /12



172.16.0.0 /12 is a designated private IP address block



65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

How many you need

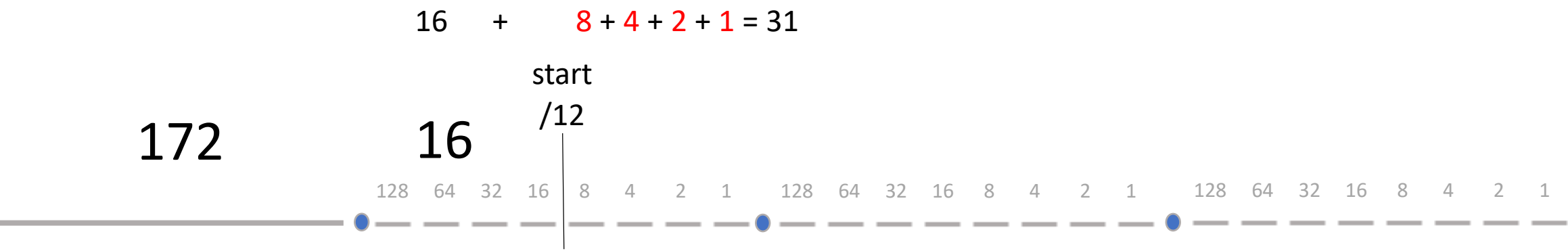
Bits to move the mask

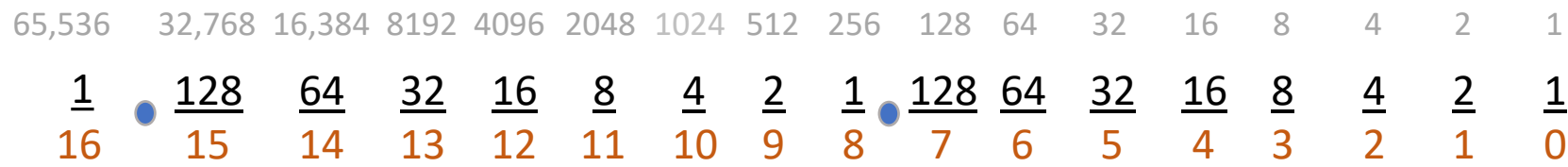
172.16.0.0 /12 is a designated private IP address block

What is the very first legitimate host IP in this block? 172.16.0.1

What is the very last legitimate host IP in this block? 172.31.255.254

Why 31? You must add the remaining bits in the octet to the starting number of 16





How many you need

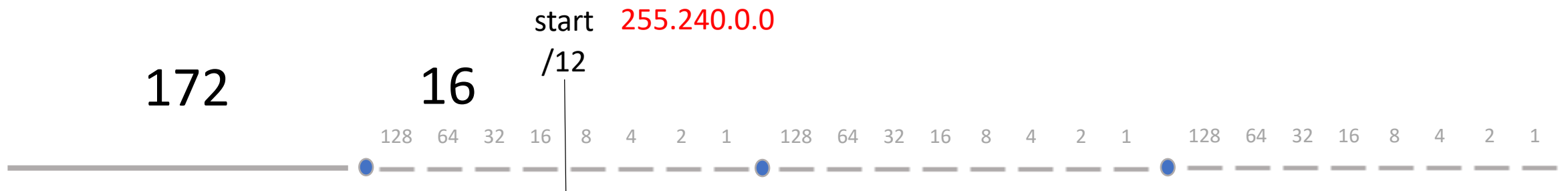
Bits to move the mask

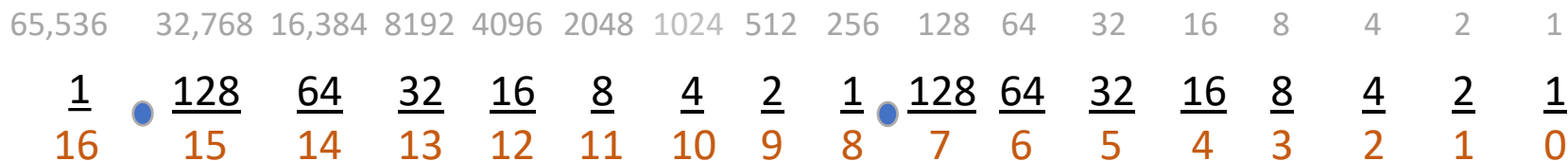
172.16.0.0 /12 is a designated private IP address block

What is the very first legitimate host IP in this block? **172.16.0.1**

What is the very last legitimate host IP in this block? **172.31.255.254**

Within this private block:





How many you need

Bits to move the mask

172.16.0.0 /12 is a designated private IP address block

What is the very first legitimate host IP in this block? **172.16.0.1**

What is the very last legitimate host IP in this block? **172.31.255.254**

Within this private block:

What is the subnet ID of this network? **172.16.0.0 /12**

Is 172.16.0.254 a legitimate host IP? **Yes**

Is 172.16.0.255 a legitimate host IP? **Yes**

Is 172.16.1.0 a legitimate host IP? **Yes**

Is 172.16.1.255 a legitimate host IP? **Yes**

Is 172.17.0.1 a legitimate host IP? **Yes**

Is 172.32.0.1 a legitimate host IP? **No – it is out of range**

Is 172.16.2.0 a legitimate host IP? **Yes**

Is 172.16.2.255 a legitimate host IP? **Yes**

Is 172.16.7.0 a legitimate host IP? **Yes**

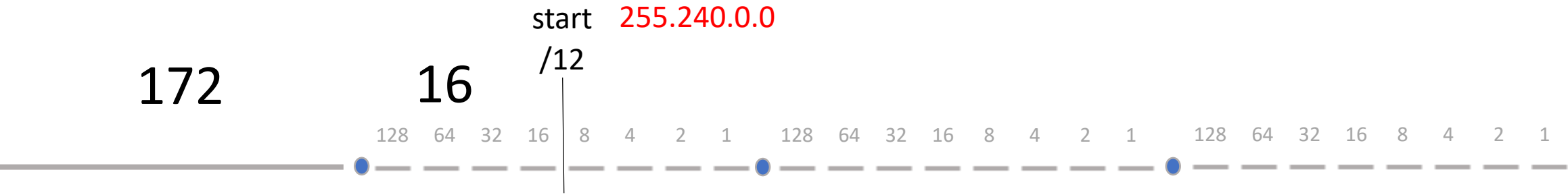
Is 172.16.7.255 a legitimate host IP? **Yes**

Is 172.31.0.255 a legitimate host IP? **Yes**

Is 172.31.255.0 a legitimate host IP? **Yes**

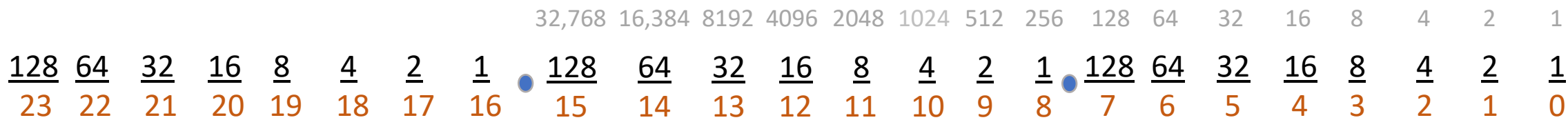
Is 172.31.255.254 a legitimate host IP? **Yes**

Is 172.31.255.255 a legitimate host IP? **No, it is the broadcast**

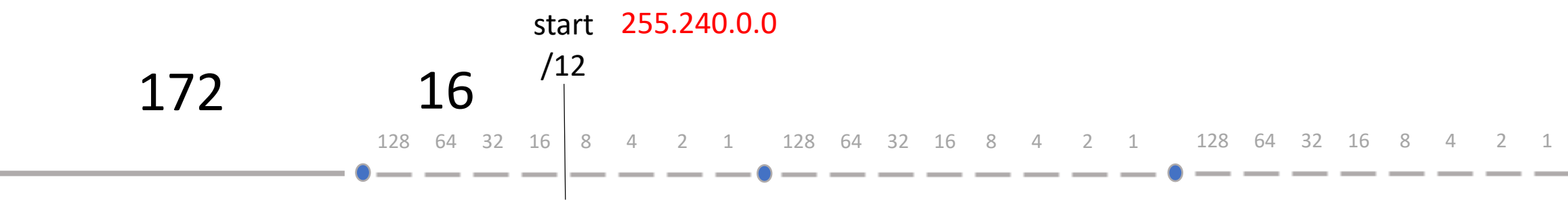


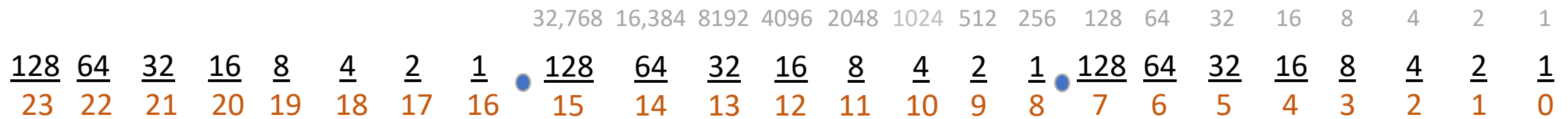
172.16.0.0 /12

Need 16 subnets



172.16.0.0 /12 (255.240.0.0)





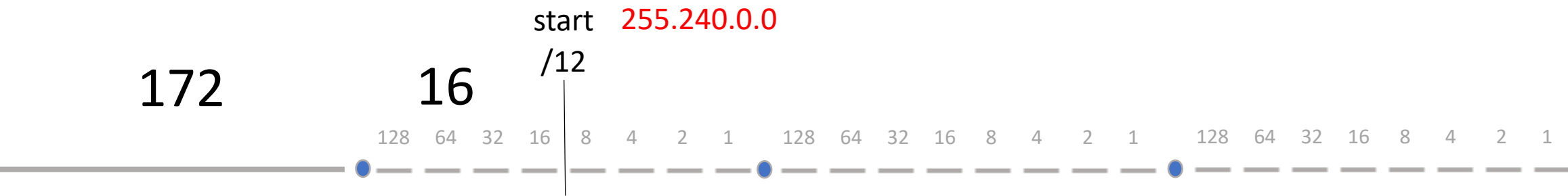
172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12



<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

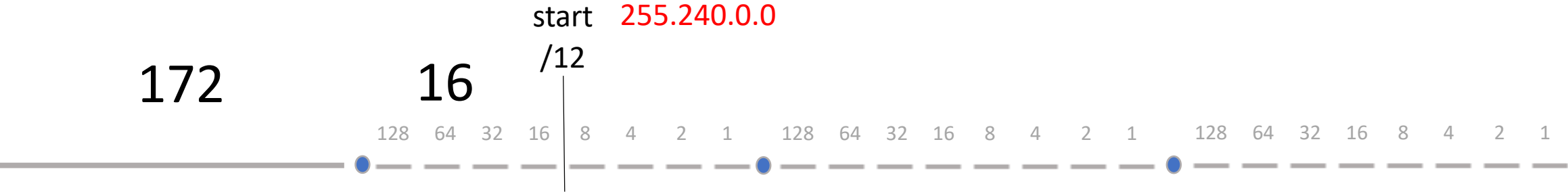
172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12



<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

172.16.0.0 /12 (255.240.0.0)

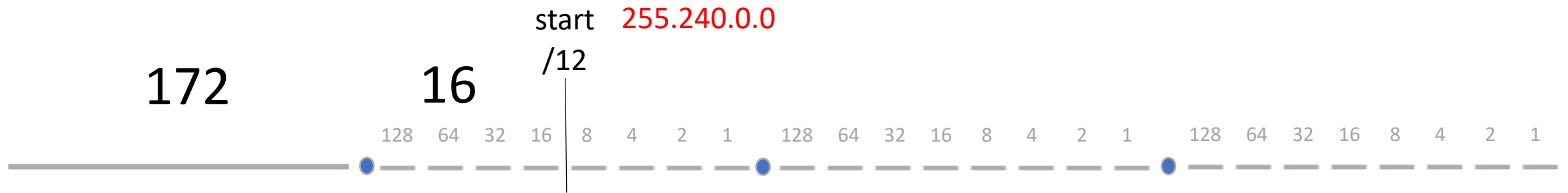
You need 16 subnets

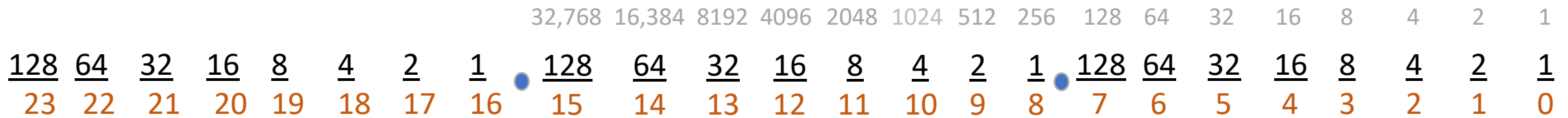
What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12





172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

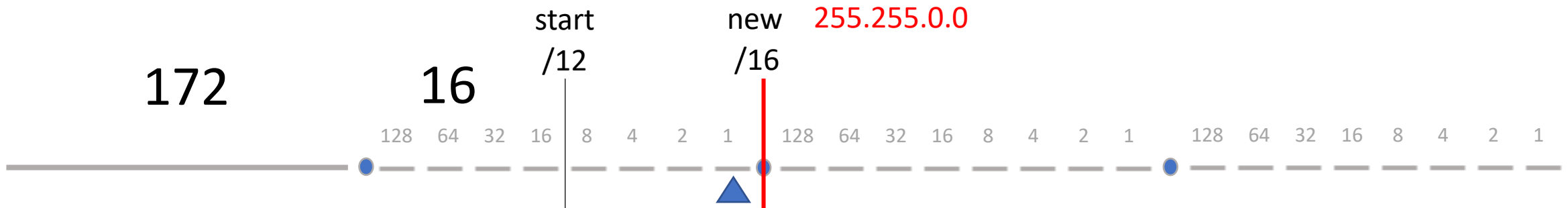
What is the broadcast? 172.31.255.255 /12

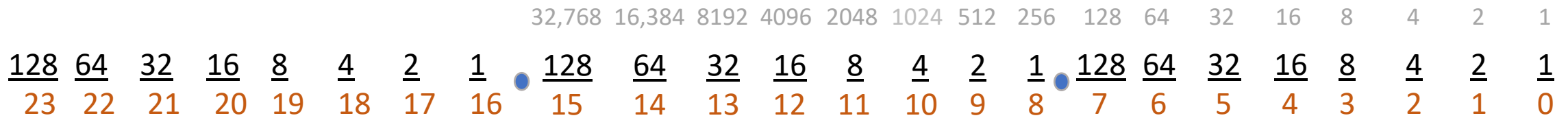
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?





172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

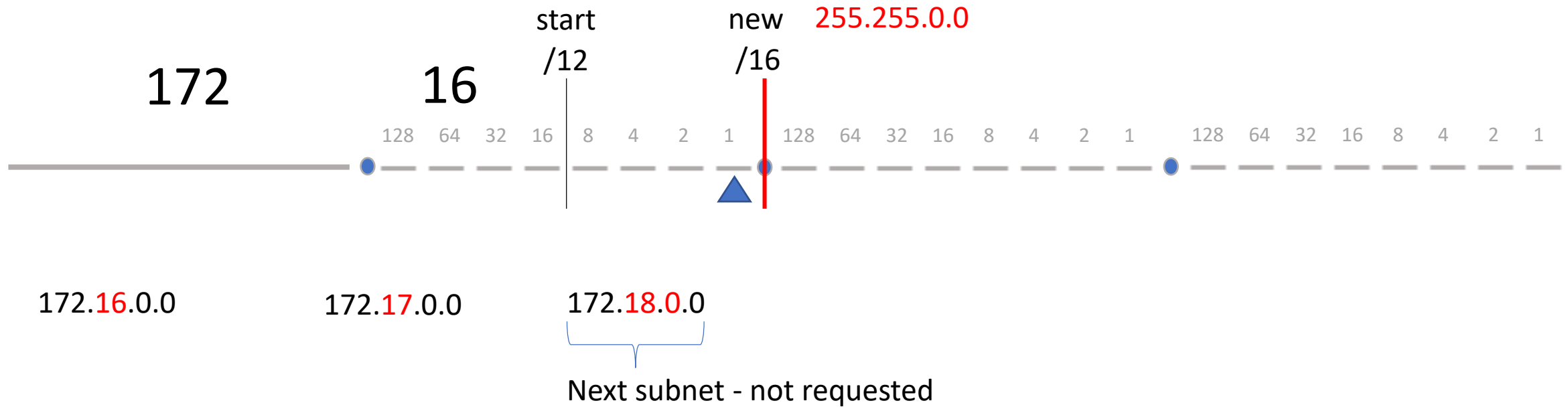
What is the broadcast? 172.31.255.255 /12

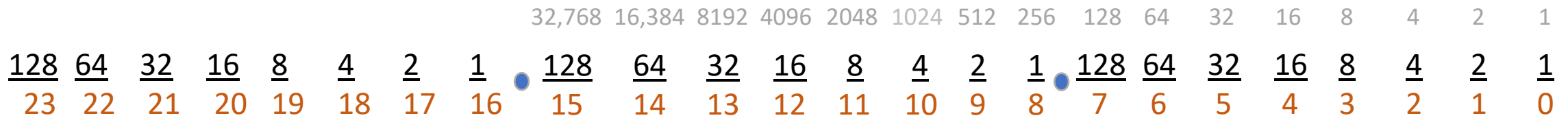
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?





172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12

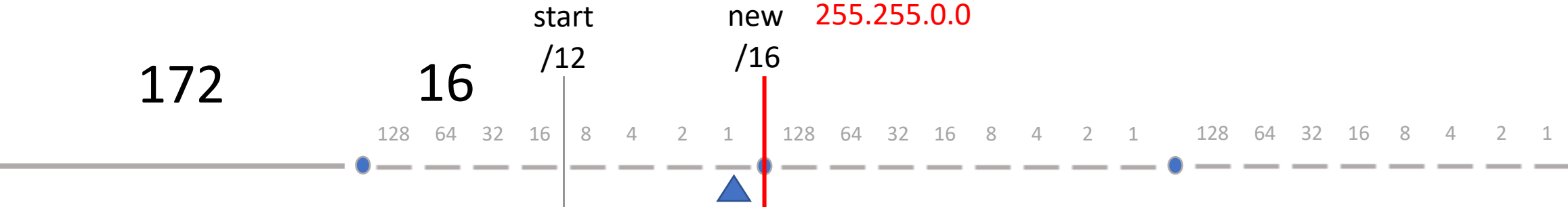
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?

What are the addresses in the last two subnets?



172.16.0.0

172.17.0.0

172.16.0.1

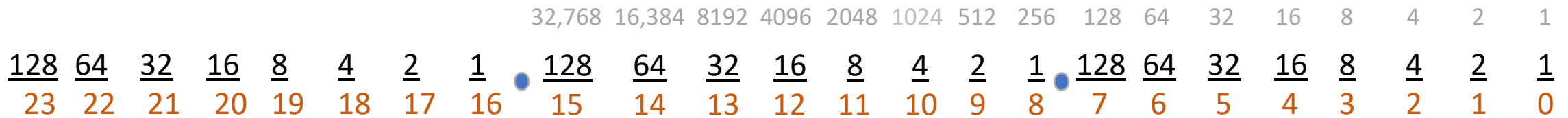
172.17.1.1

172.16.255.254

172.17.255.254

172.16.255.255

172.17.255.255



172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12

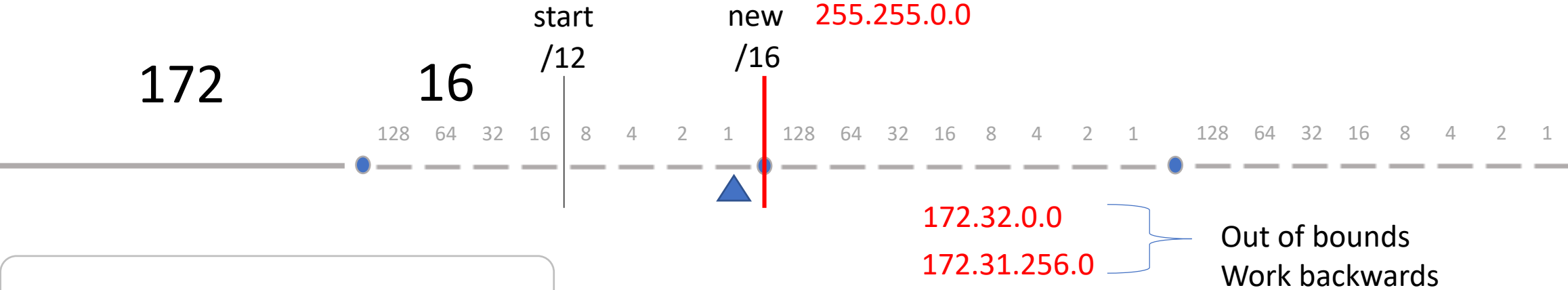
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

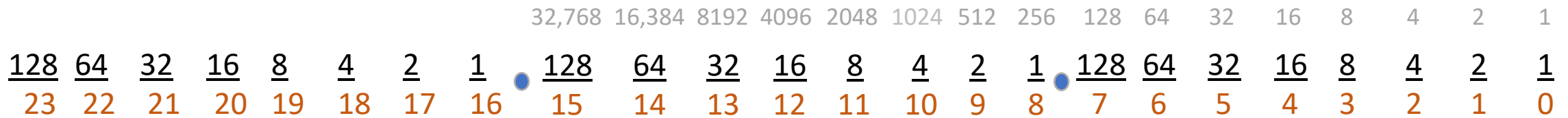
What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?

What are the addresses in the last two subnets?



172.16.0.0	172.17.0.0
172.16.0.1	172.17.1.1
172.16.255.254	172.17.255.254
172.16.255.255	172.17.255.255



172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12

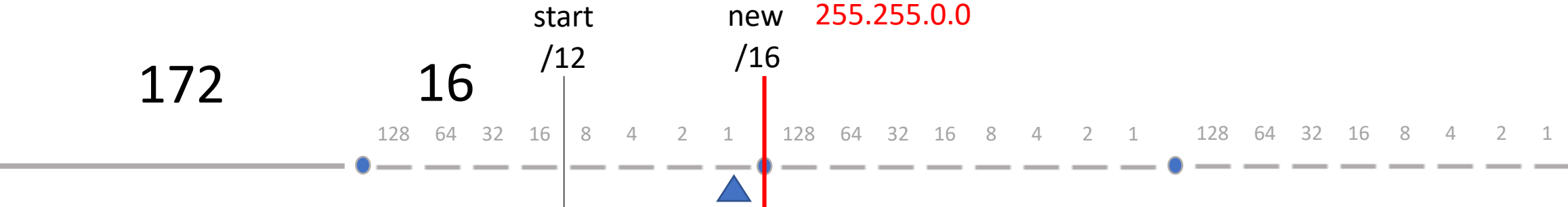
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?

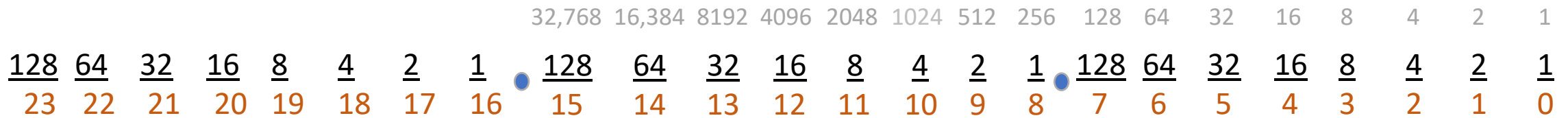
What are the addresses in the last two subnets?



172.16.0.0	172.16.1.0
172.16.0.1	172.16.1.1
172.16.0.254	172.16.1.254
172.16.0.255	172.16.1.255

172.30.0.0 172.31.0.0

172.32.0.0



172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12

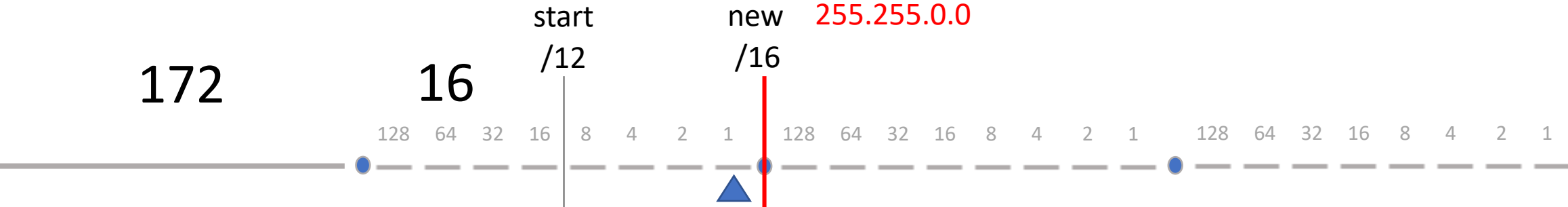
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?

What are the addresses in the last two subnets?



172.16.0.0

172.16.1.0

172.16.0.1

172.16.1.1

172.16.0.254

172.16.1.254

172.16.0.255

172.16.1.255

172.30.0.0

172.31.0.0

172.30.0.1

172.31.0.1

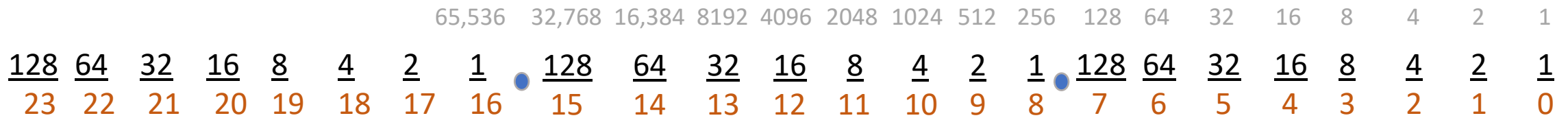
172.30.255.254

172.31.255.254

172.30.255.255

172.31.255.255

172.32.0.0



172.16.0.0 /12 (255.240.0.0)

What is the network ID? 172.16.0.0 /12

What is the first host IP? 172.16.0.1 /12

What is the last host IP? 172.31.255.254 /12

What is the broadcast? 172.31.255.255 /12

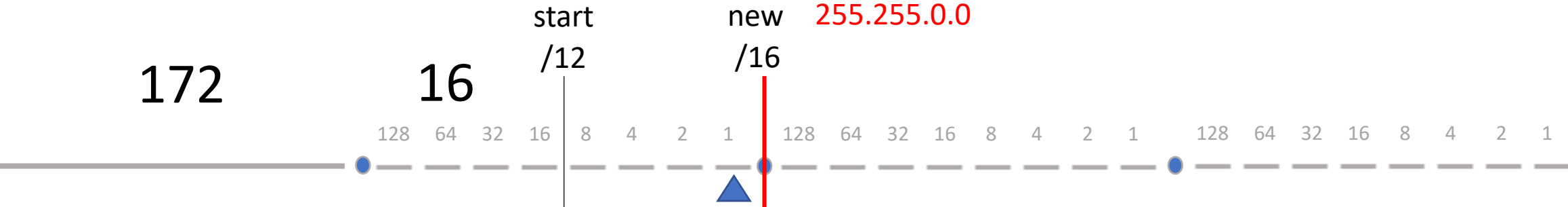
You need 16 subnets

What is the new subnet mask? /16 (255.255.0.0)

What is the delta, and in which octet? 1, 2nd

What are the addresses in the first two subnets?

What are the addresses in the last two subnets?



172.16.0.0

172.16.1.0

172.16.0.1

172.16.1.1

172.16.0.254

172.16.1.254

172.16.0.255

172.16.1.255

172.30.0.0

172.31.0.0

172.30.0.1

172.31.0.1

172.30.0.254

172.31.255.254

172.30.0.255

172.31.255.255

172.17.0.0 / 16

Need 1000 subnets

65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

Start – 172.17.0.0 / 16
 What is the full range of host IP addresses?



65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

Start – 172.17.0.0 / 16

What is the full range of host IP addresses? 172.17.0.1 – 172.17.255.254



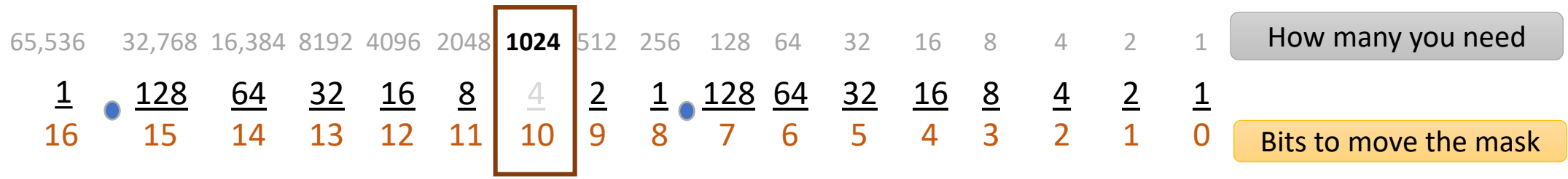
65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

Start – 172.17.0.0 / 16
 What is the full range of host IP addresses?
 Need 1000 subnets





Start – 172.17.0.0 / 16

What is the full range of host IP addresses?

Need 1000 subnets

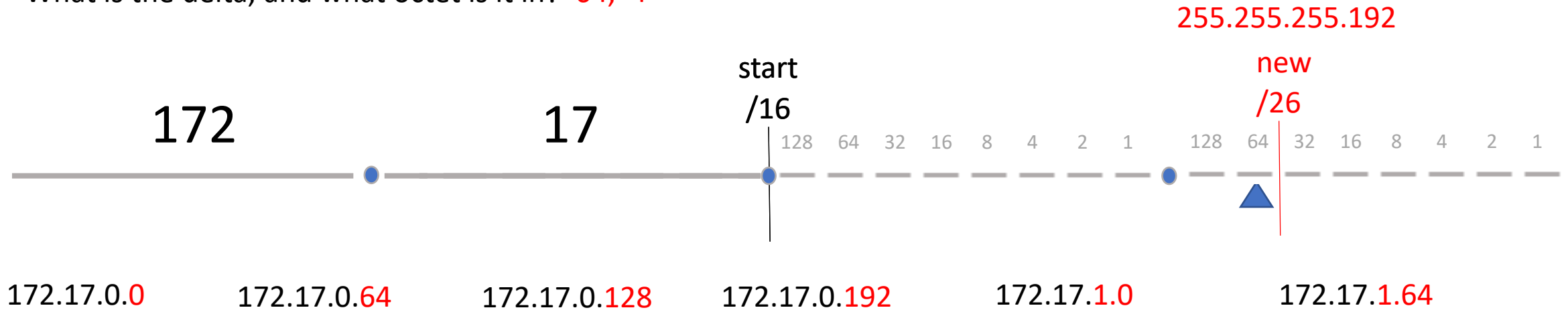
What is the new subnet mask? /26 (255.255.255.192)

What is the delta, and what octet is it in? 64, 4th

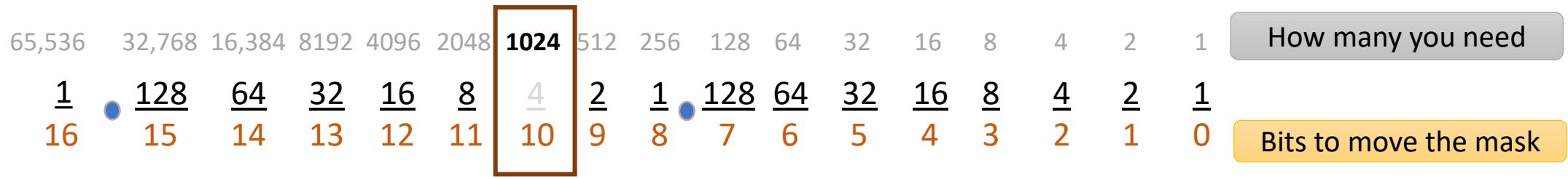
With the new mask, how many hosts per subnet?

$$2^6 - 2 = 62$$

What are the subnet IDs for the first 6 subnets?



172.17.1.128



Start – 172.17.0.0 / 16

What is the full range of host IP addresses?

Need 1000 subnets

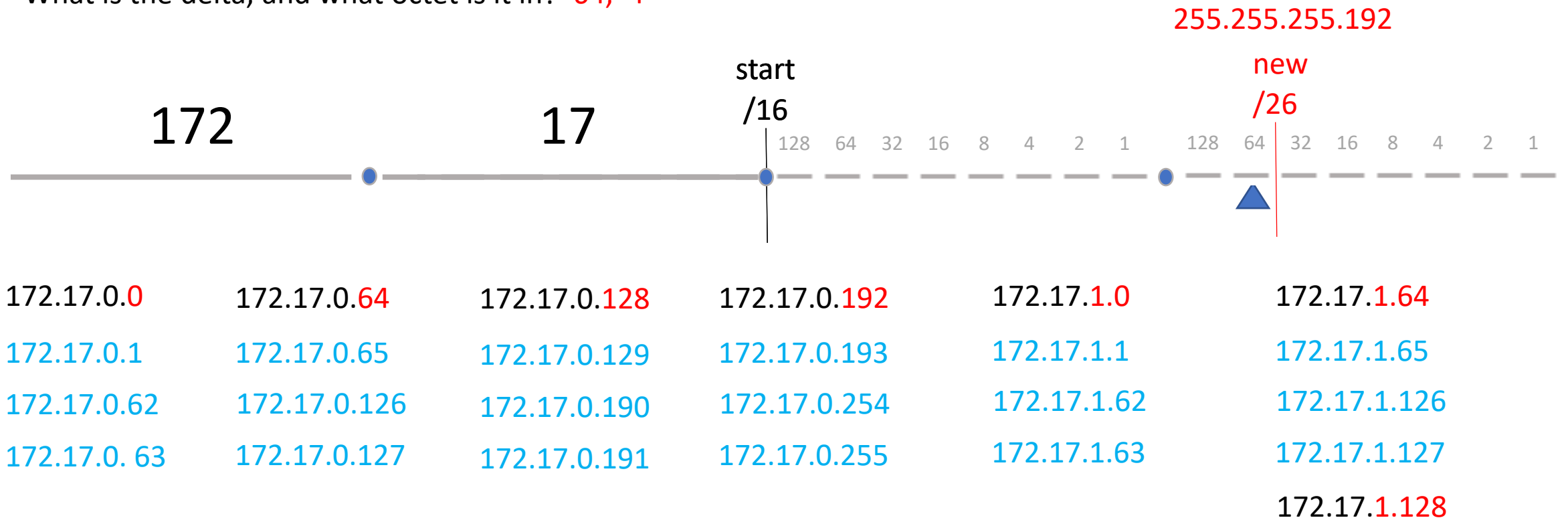
What is the new subnet mask? /26 (255.255.255.192)

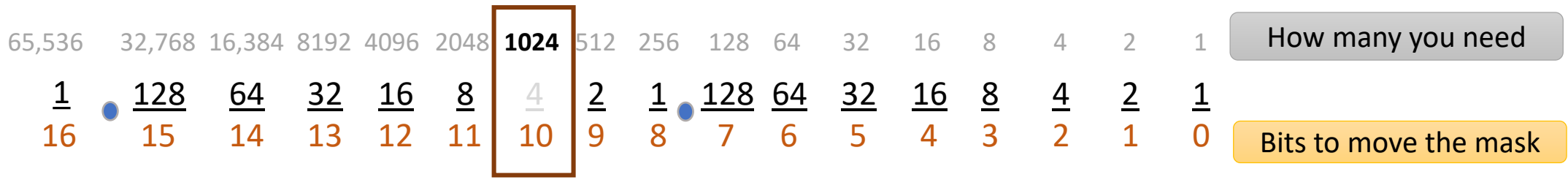
What is the delta, and what octet is it in? 64, 4th

With the new mask, how many hosts per subnet?

$$2^6 - 2 = 62$$

What are the subnet IDs for the first 6 subnets?





Start – 172.17.0.0 / 16

What is the full range of host IP addresses?

Need 1000 subnets

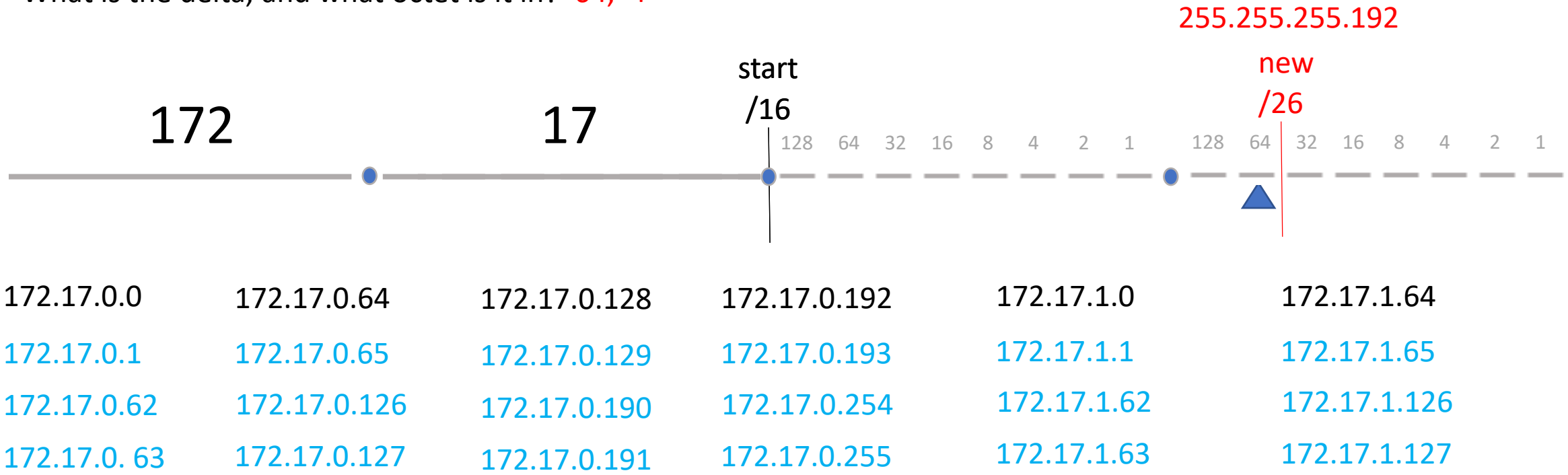
What is the new subnet mask? **/26 (255.255.255.192)**

What is the delta, and what octet is it in? **64, 4th**

With the new mask, how many hosts per subnet?

$2^6 - 2 = 62$

What are the subnet IDs for the first 6 subnets?



10.0.0.0 / 8

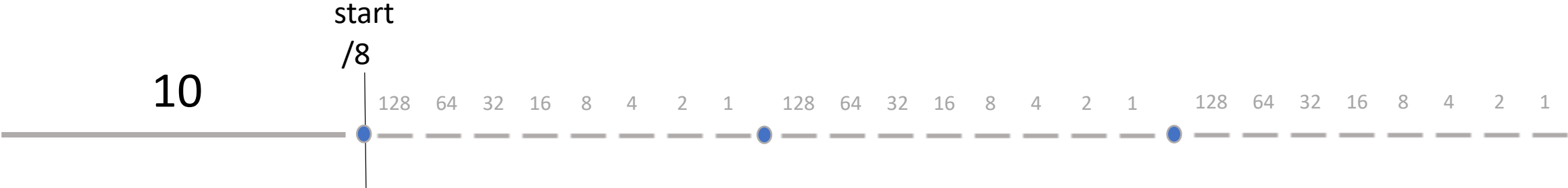
Need 30 subnets

Start – 10.0.0.0 / 8

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

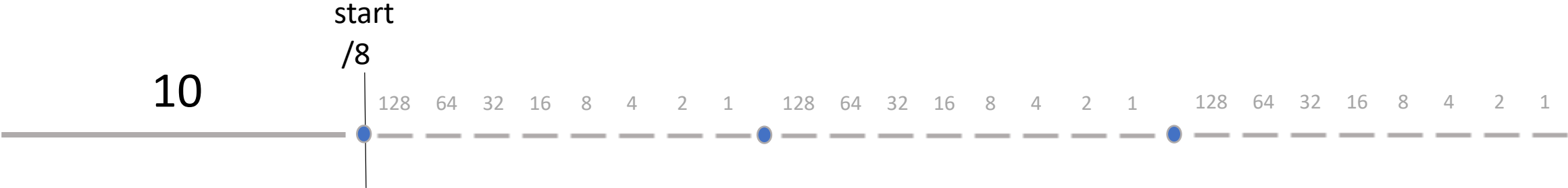


Start – 10.0.0.0 / 8
What is the full range of host IPs?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



Start – 10.0.0.0 / 8

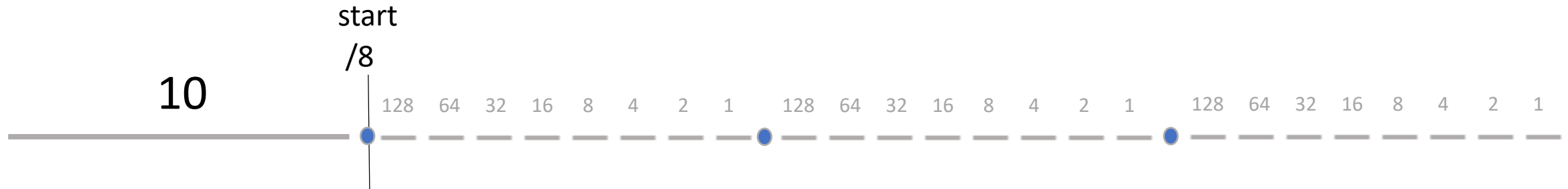
What is the full range of host IPs?

10.0.0.1 – 10.255.255.254

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



Start – 10.0.0.0 / 8

What is the full range of host IPs?

Need 30 subnets

128
7

64
6

32
5

16
4

8
3

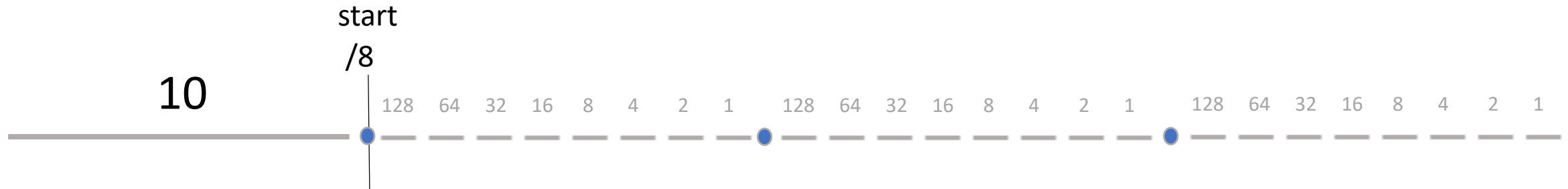
4
2

2
1

1
0

How many you need

Bits to move the mask



Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? **/13 (255.248.0.0)**

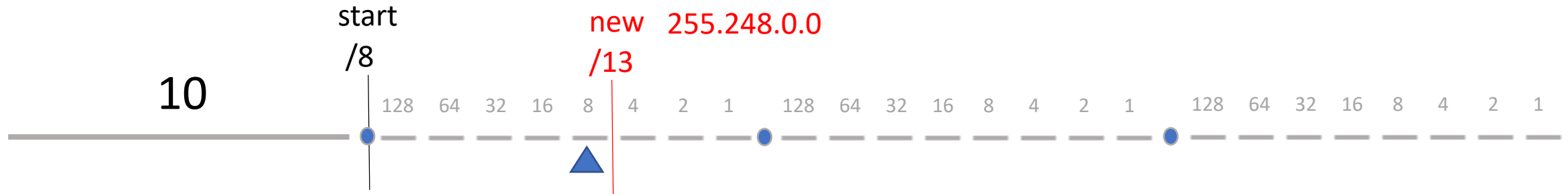
What is the delta, and in which octet? **8, 2nd**

With the new mask, how many hosts per subnet? **$2^{19} - 2 = 524,286$**

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



What are the first two subnets?

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? **/13 (255.248.0.0)**

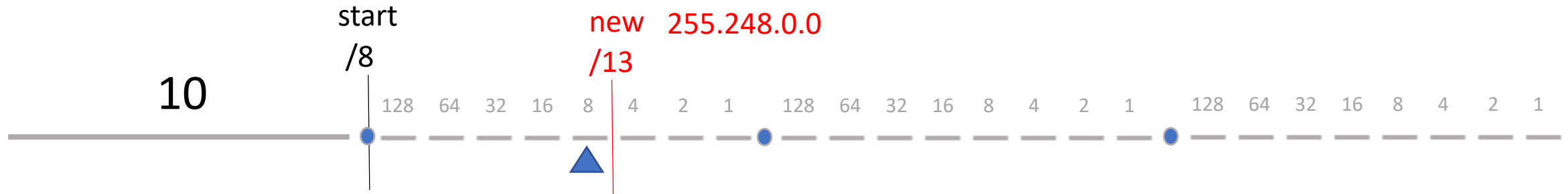
What is the delta, and in which octet? **8, 2nd**

With the new mask, how many hosts per subnet? **$2^{19} - 2 = 524,286$**

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

10.0.0.0

10.8.0.0

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

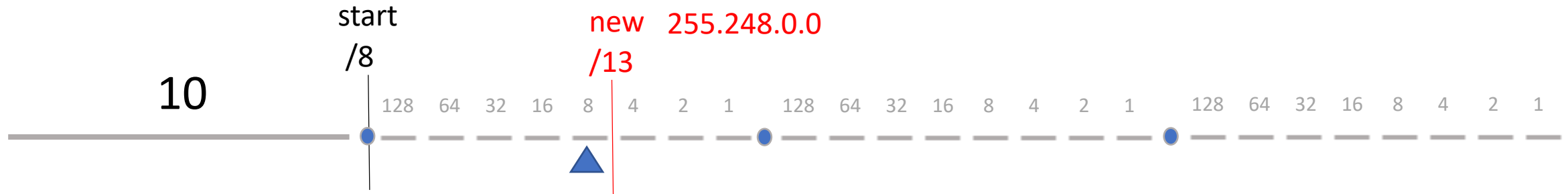
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

10.0.0.0

10.0.0.1

10.7.255.254

10.7.255.255

10.8.0.0

10.8.0.1

10.15.255.254

10.15.255.255

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

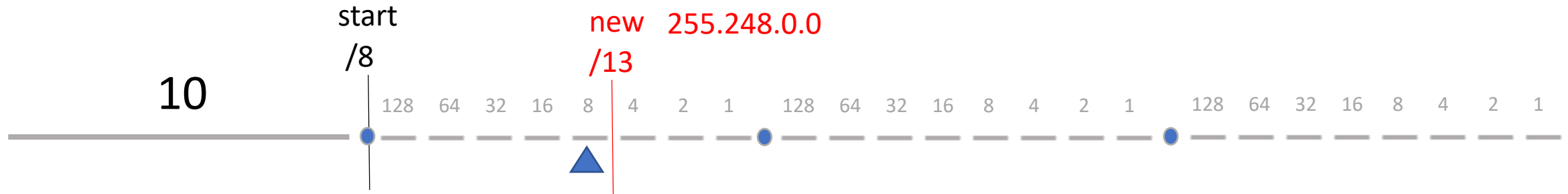
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

10.0.0.0
10.0.0.1
10.7.255.254
10.7.255.255
10.8.0.0
10.8.0.1
10.15.255.254
10.15.255.255

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

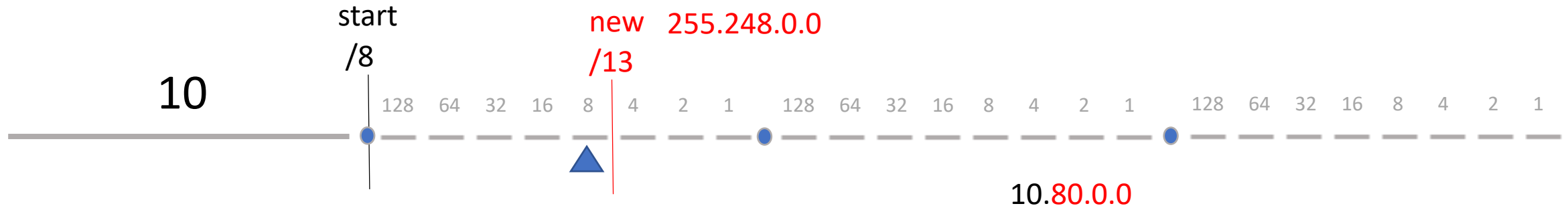
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

10.0.0.0
10.0.0.1
10.7.255.254
10.7.255.255
10.8.0.0
10.8.0.1
10.15.255.254
10.15.255.255

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

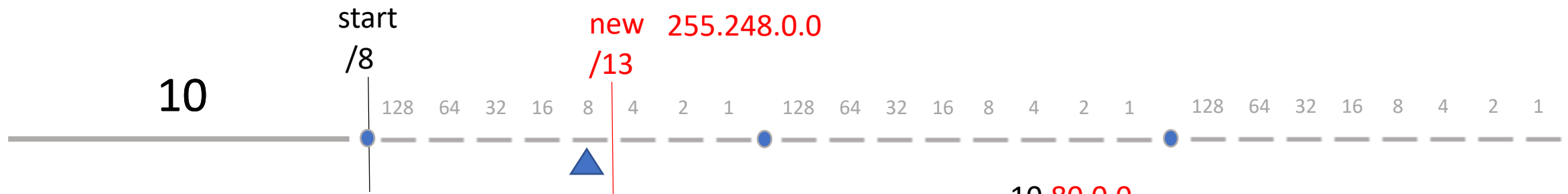
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

10.0.0.0

10.0.0.1

10.7.255.254

10.7.255.255

10.8.0.0

10.8.0.1

10.15.255.254

10.15.255.255

10.80.0.0

10.88.0.0

10.96.0.0

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

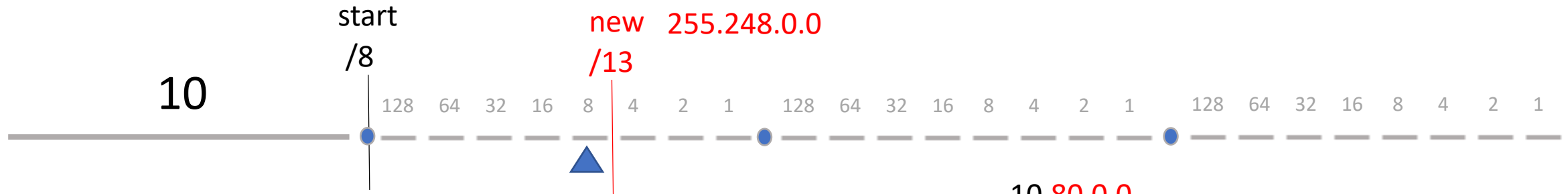
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

128	64	32	16	8	4	2	1
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

10.0.0.0
10.0.0.1
10.7.255.254
10.7.255.255
10.8.0.0
10.8.0.1
10.15.255.254
10.15.255.255

10.80.0.0
10.88.0.0
10.88.0.1
10.95.255.254
10.95.255.255
10.96.0.0
10.96.0.1
10.103.255.254
10.103.255.255
10.104.0.0

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? **/13 (255.248.0.0)**

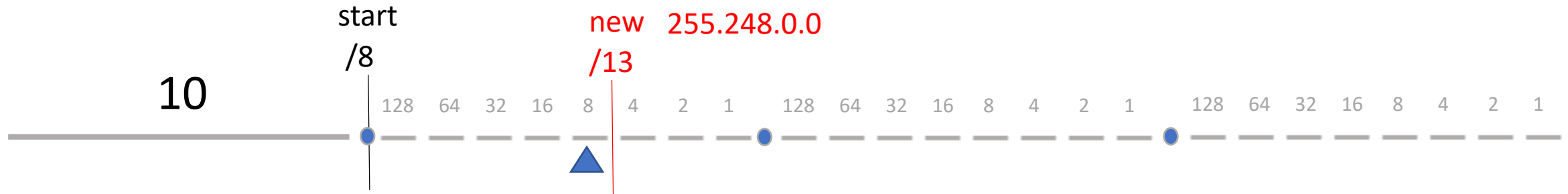
What is the delta, and in which octet? **8, 2nd**

With the new mask, how many hosts per subnet? **$2^{19} - 2 = 524,286$**

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

What are the last two subnets?

10.0.0.0

10.0.0.1

10.7.255.254

10.7.255.255

10.8.0.0

10.8.0.1

10.15.255.254

10.15.255.255

10.88.0.0

10.88.0.1

10.95.255.254

10.95.255.255

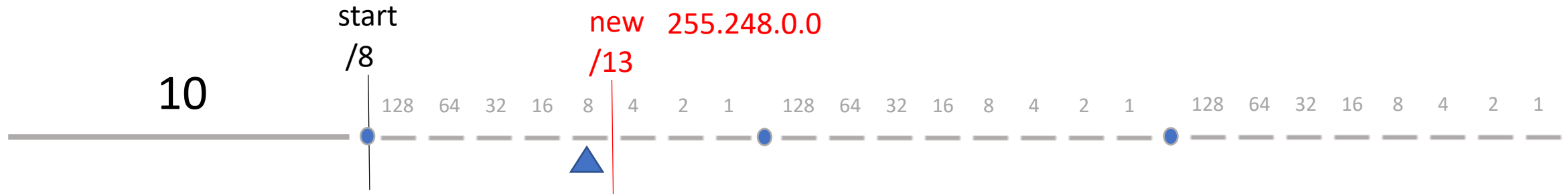
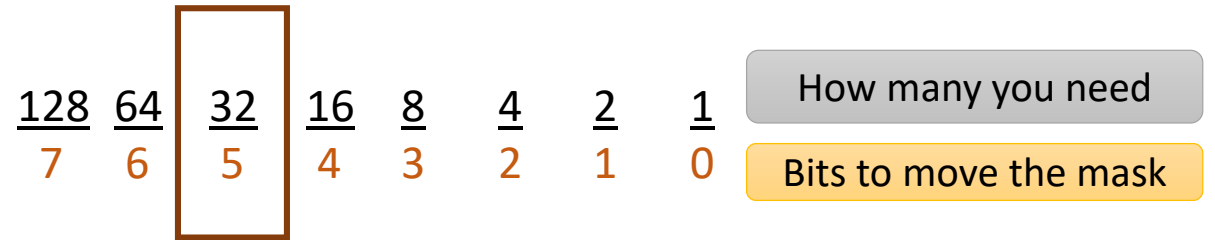
10.96.0.0

10.96.0.1

10.103.255.254

10.103.255.255

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$



What are the last two subnets?

10.103.255.255

10.256.0.0
11.0.0.0

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

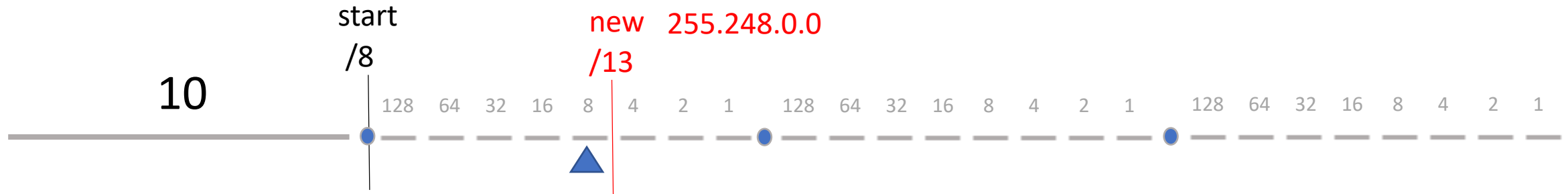
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

What are the last two subnets?

10.0.0.0

10.0.0.1

10.7.255.254

10.7.255.255

10.8.0.0

10.8.0.1

10.15.255.254

10.15.255.255

10.88.0.0

10.88.0.1

10.95.255.254

10.95.255.255

10.96.0.0

10.96.0.1

10.103.255.254

10.103.255.255

10.240.0.0

10.248.0.0

10.256.0.0

11.0.0.0

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

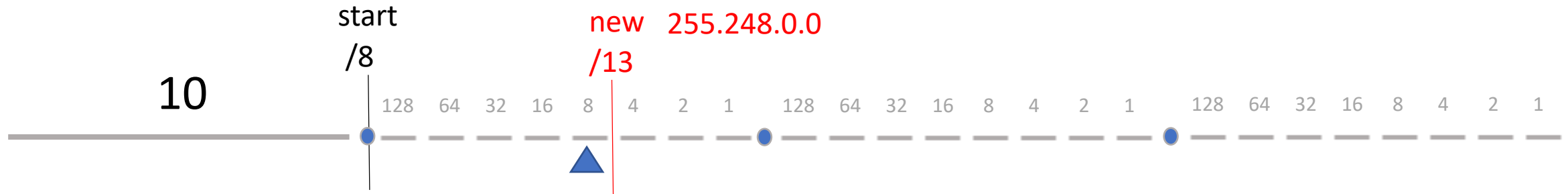
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

$\frac{128}{7}$	$\frac{64}{6}$	$\frac{32}{5}$	$\frac{16}{4}$	$\frac{8}{3}$	$\frac{4}{2}$	$\frac{2}{1}$	$\frac{1}{0}$
-----------------	----------------	----------------	----------------	---------------	---------------	---------------	---------------

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

What are the last two subnets?

10.0.0.0

10.0.0.1

10.7.255.254

10.7.255.255

10.8.0.0

10.8.0.1

10.15.255.254

10.15.255.255

10.88.0.0

10.88.0.1

10.95.255.254

10.95.255.255

10.96.0.0

10.96.0.1

10.103.255.254

10.103.255.255

10.240.0.0

10.240.0.1

10.247.255.254

10.247.255.255

10.248.0.0

10.248.0.1

10.255.255.254

10.255.255.255

10.256.0.0

11.0.0.0

Start – 10.0.0.0 / 8

What is the full range of IP?

Need 30 subnets

What is the new subnet mask? /13 (255.248.0.0)

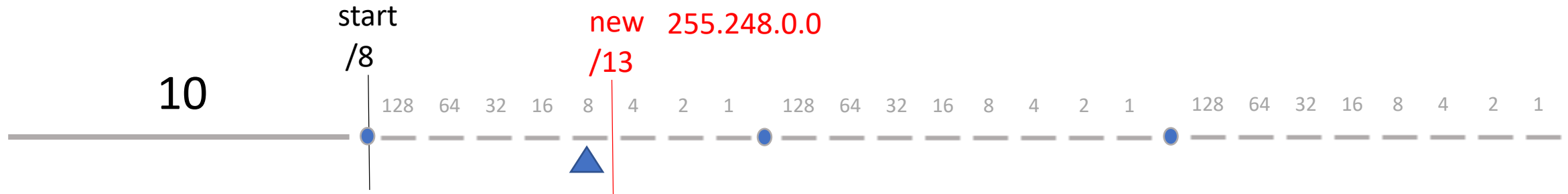
What is the delta, and in which octet? 8, 2nd

With the new mask, how many hosts per subnet? $2^{19} - 2 = 524,286$

128	64	32	16	8	4	2	1
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask



What are the first two subnets?

What are the next two subnets after 10.80.0.0?

What are the last two subnets?

10.0.0.0

10.0.0.1

10.7.255.254

10.7.255.255

10.8.0.0

10.8.0.1

10.15.255.254

10.15.255.255

10.88.0.0

10.88.0.1

10.95.255.254

10.95.255.255

10.96.0.0

10.96.0.1

10.103.255.254

10.103.255.255

10.240.0.0

10.240.0.1

10.247.255.254

10.247.255.255

10.248.0.0

10.248.0.1

10.255.255.254

10.255.255.255

10.0.0.0 / 8

Need 10,000 subnets

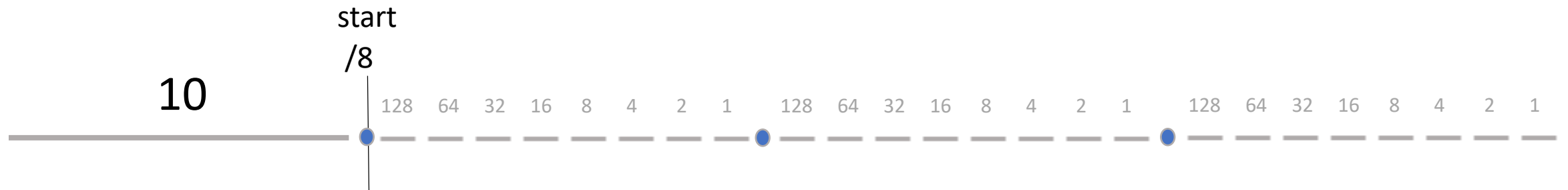
65,536 32,768 16,384 8192 4096 2048 1024 512 256 128 64 32 16 8 4 2 1

How many you need

1 128 64 32 16 8 4 2 1 128 64 32 16 8 4 2 1
16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bits to move the mask

Start – 10.0.0.0 / 8



65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

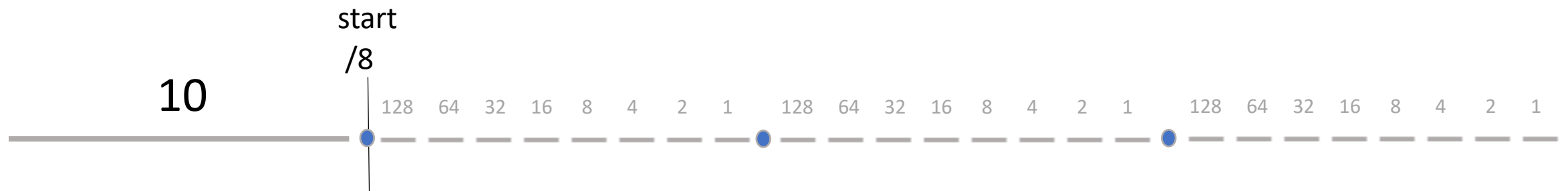
How many you need

Bits to move the mask

Start – 10.0.0.0 / 8

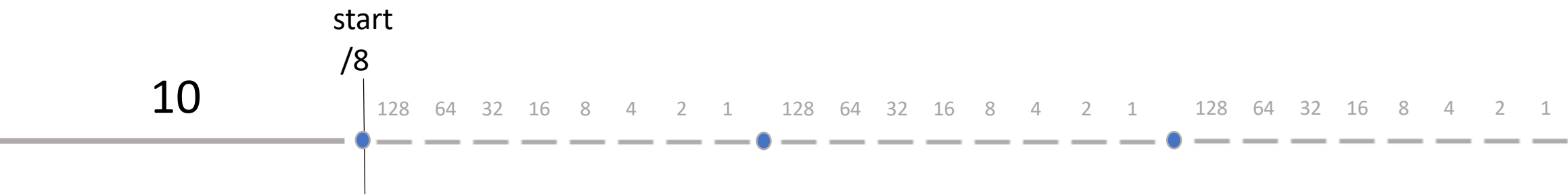
What is the subnet ID, first host IP, last host IP, and broadcast?

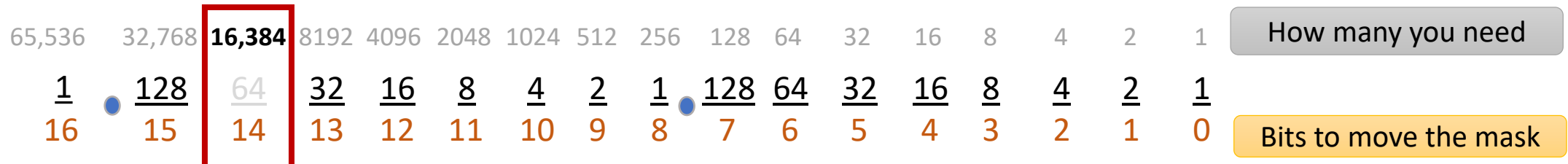
10.0.0.0 /8, 10.0.0.1, 10.255.255.254, 10.255.255.255



65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1	How many you need	
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	Bits to move the mask	
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		

Start – 10.0.0.0 / 8
 Need 10,000 subnets





Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

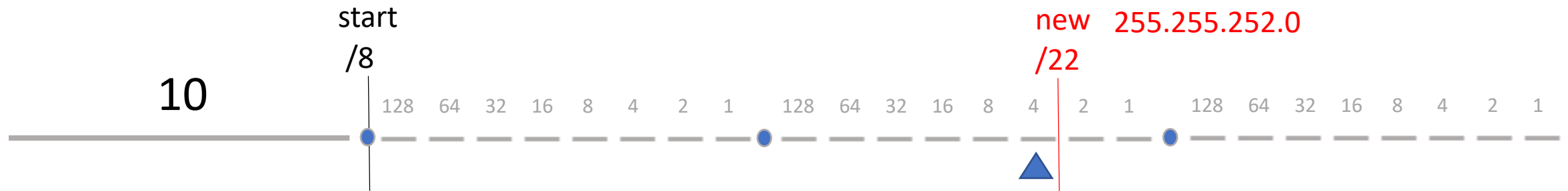
What is the delta, and in which octet? 4, 3rd

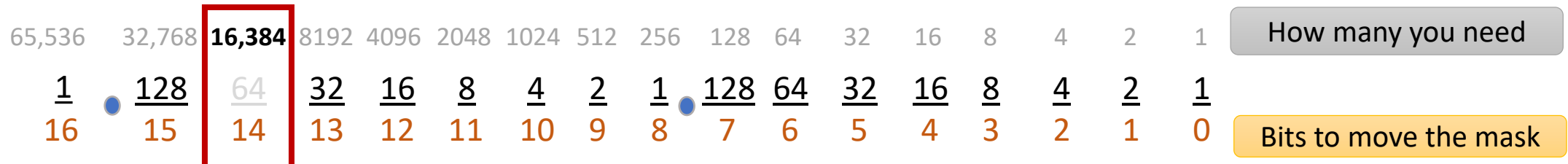
What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?





Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

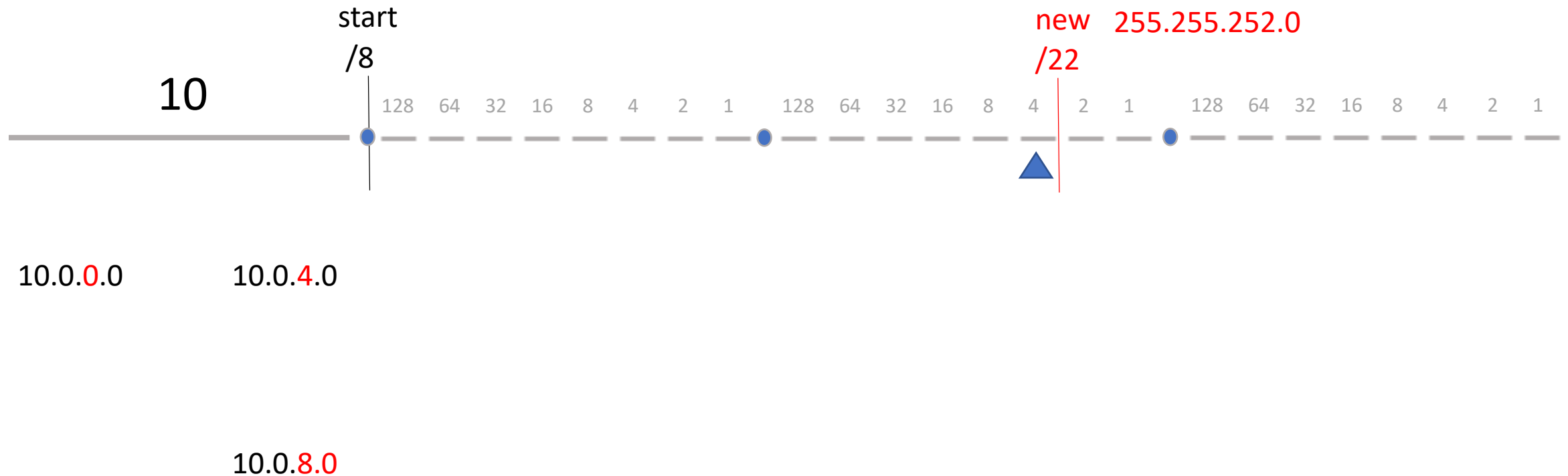
What is the delta, and in which octet? 4, 3rd

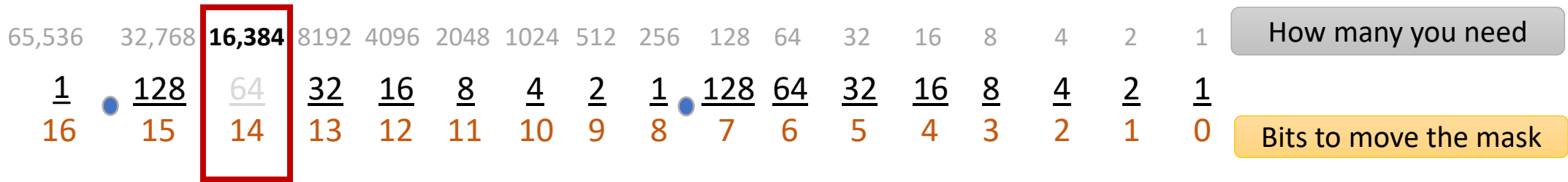
What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?





Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

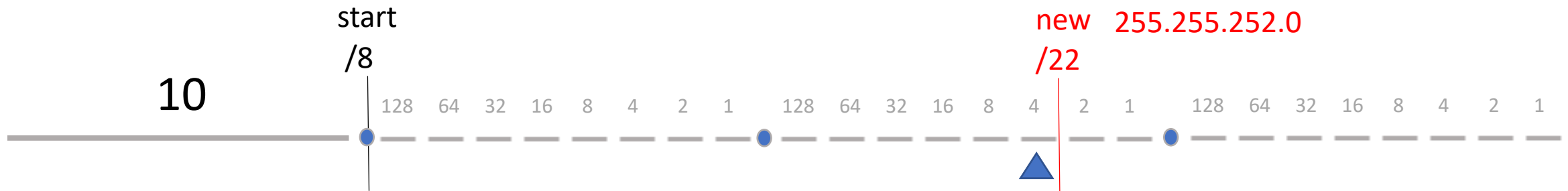
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?



10.0.0.0

10.0.4.0

10.0.0.1

10.0.4.1

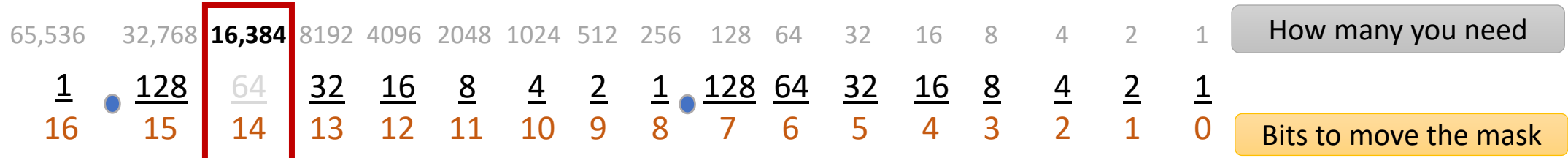
10.0.3.254

10.0.7.254

10.0.3.255

10.0.7.255

10.0.8.0



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? **/22 (255.255.252.0)**

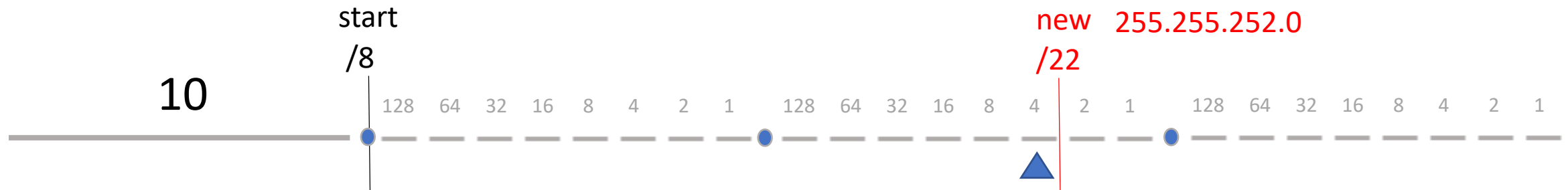
What is the delta, and in which octet? **4, 3rd**

What is the subnet ID for the first two subnets?

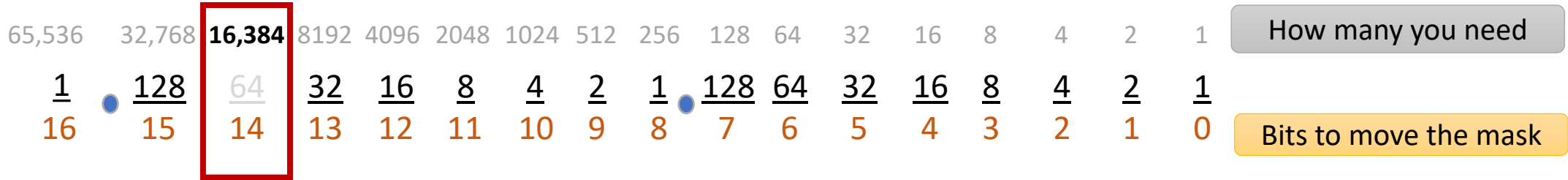
What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?

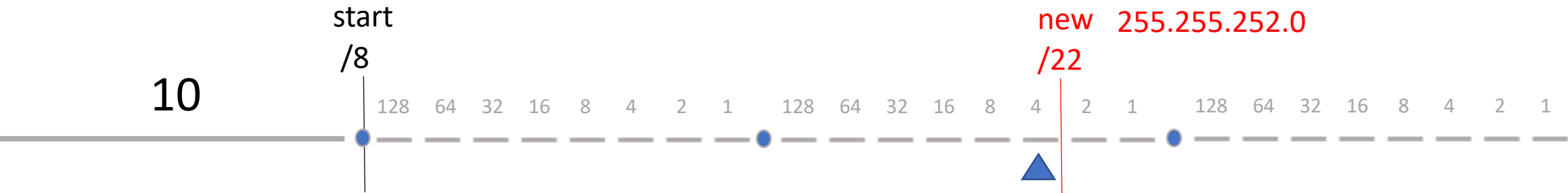


10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255



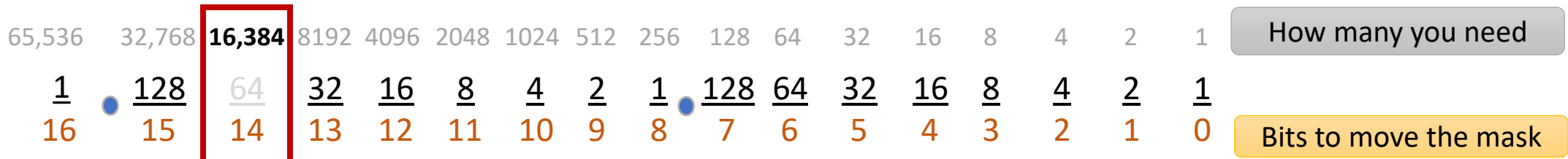
Start – 10.0.0.0 / 8
Need 10,000 subnets
What is the new subnet mask? /22 (255.255.252.0)
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?
What are the next two subnets after 10.0.40.0?
What are the next two subnets after 10.40.0.0?
What is the last subnet ID?



10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0.40.0
10.0.44.0
10.0.48.0



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

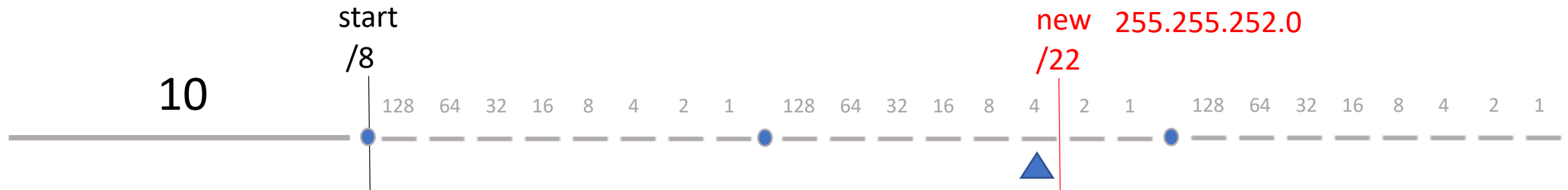
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?



10.0.0.0

10.0.4.0

10.0.0.1

10.0.4.1

10.0.3.254

10.0.7.254

10.0.3.255

10.0.7.255

10.0.40.0

10.0.44.0

10.0.44.1

10.0.47.254

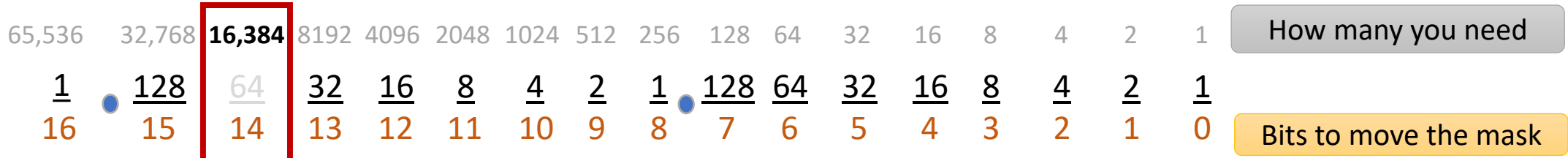
10.0.47.255

10.0.48.0

10.0.48.1

10.0.51.254

10.0.51.255



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

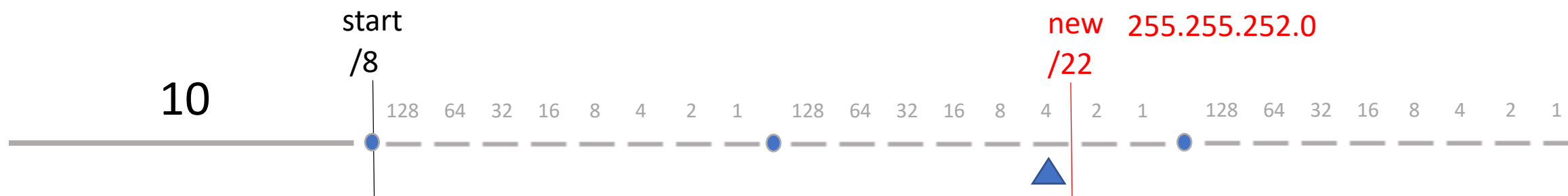
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

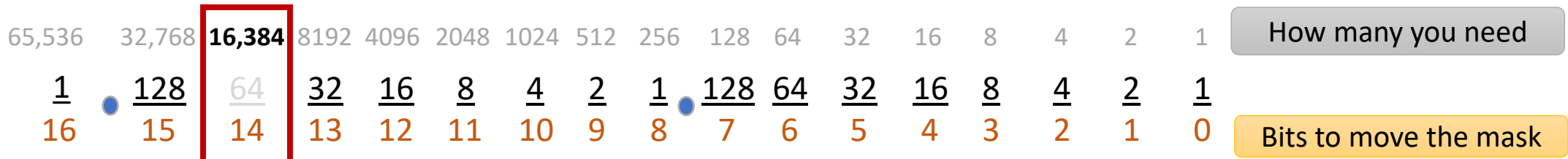
What are the next two subnets after 10.40.0.0?

What is the last subnet ID?



10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0.44.0	10.0.48.0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

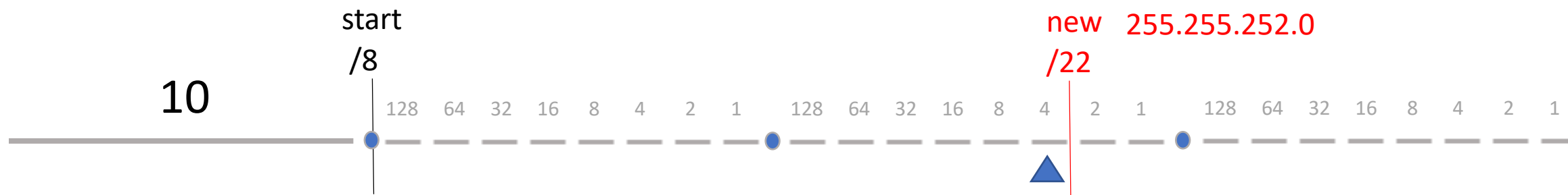
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?



10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

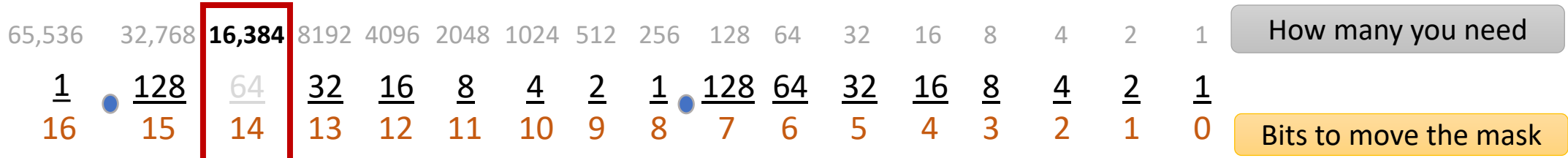
10.0.44.0	10.0.48.0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255

10.40.0.0

10.40.4.0

10.40.8.0

10.40.12.0



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

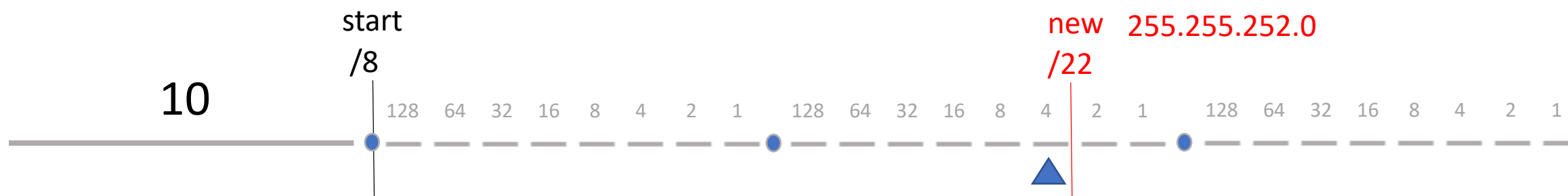
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

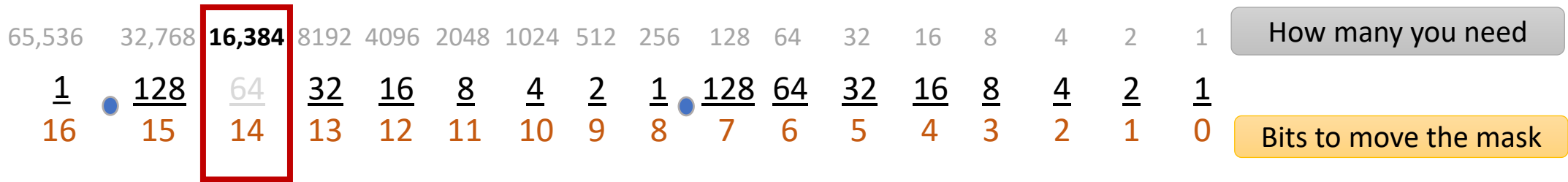
What is the last subnet ID?



10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0.44.0	10.0.48.0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255

10.40.4.0	10.40.8.0
10.40.4.1	10.40.8.1
10.40.7.254	10.40.11.254
10.40.7.255	10.40.11.255



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

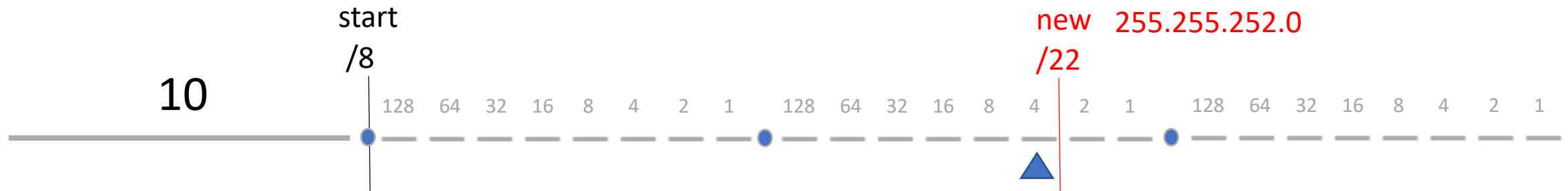
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?

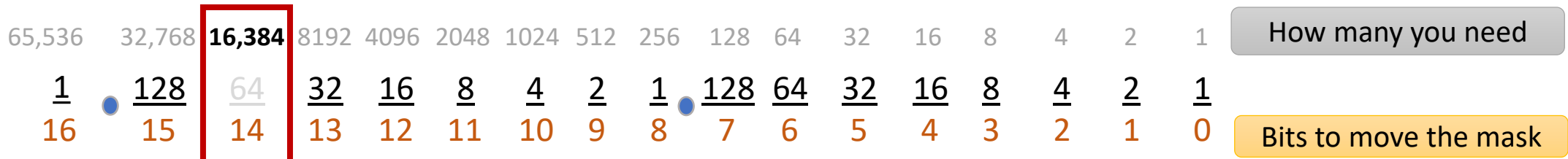


10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0.44.0	10.0.48.0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255

10.40.4.0	10.40.8.0
10.40.4.1	10.40.8.1
10.40.7.254	10.40.11.254
10.40.7.255	10.40.11.255

10.256.0.0
11.0.0.0



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? **/22 (255.255.252.0)**

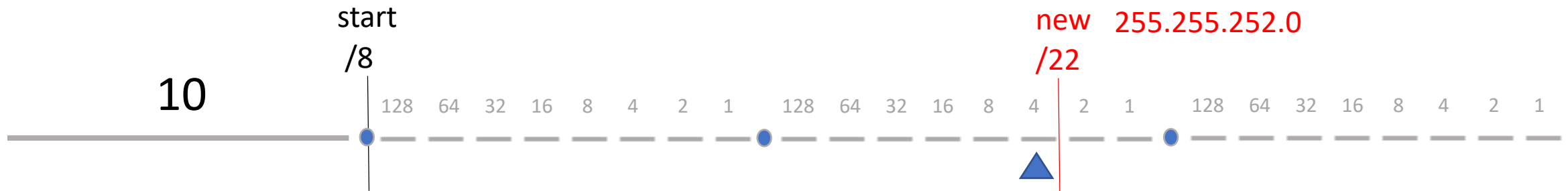
What is the delta, and in which octet? **4, 3rd**

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?

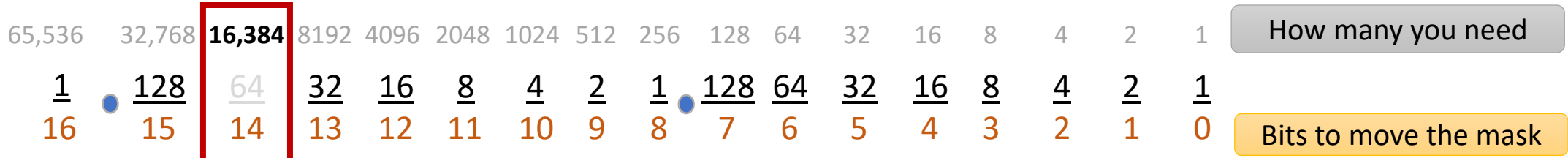


10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0. 44 .0	10.0. 48 .0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255

10.40. 4 .0	10.40. 8 .0
10.40.4.1	10.40.8.1
10.40.7.254	10.40.11.254
10.40.7.255	10.40.11.255

11.0.**0**.0



Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

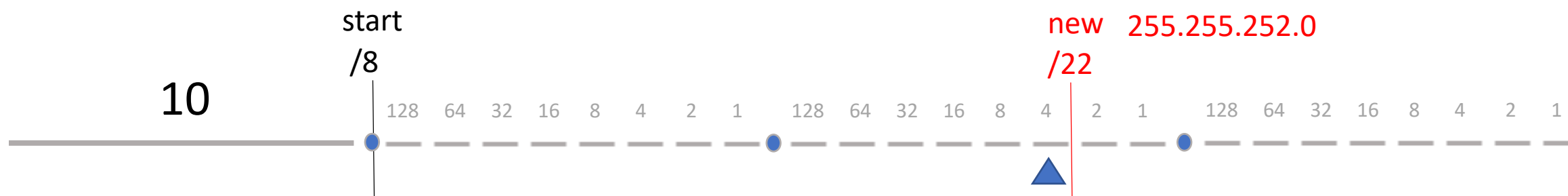
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?



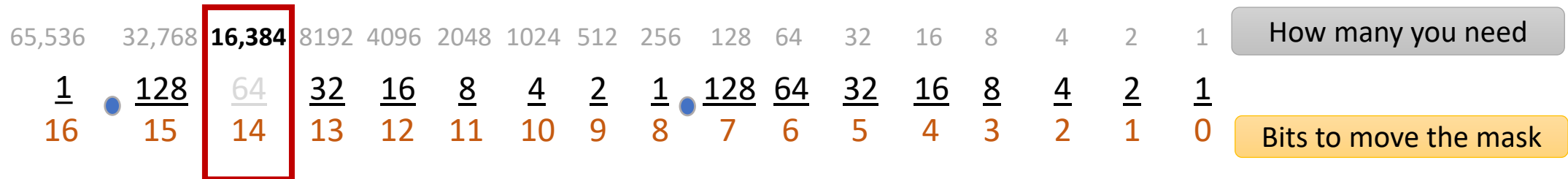
10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0.44.0	10.0.48.0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255

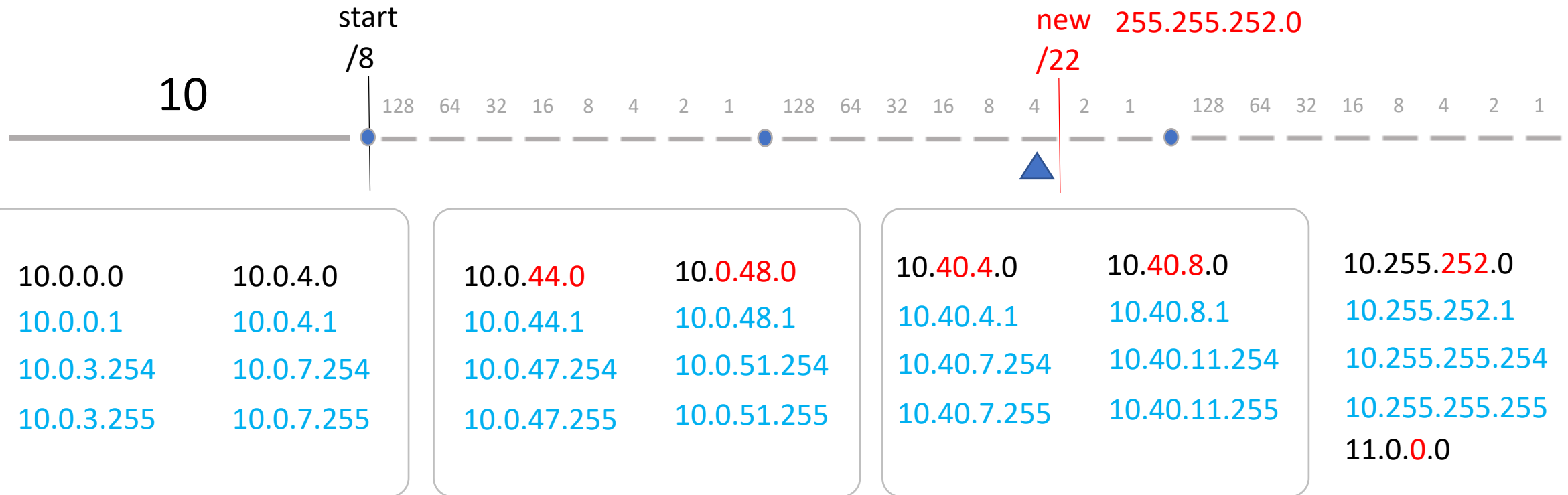
10.40.4.0	10.40.8.0
10.40.4.1	10.40.8.1
10.40.7.254	10.40.11.254
10.40.7.255	10.40.11.255

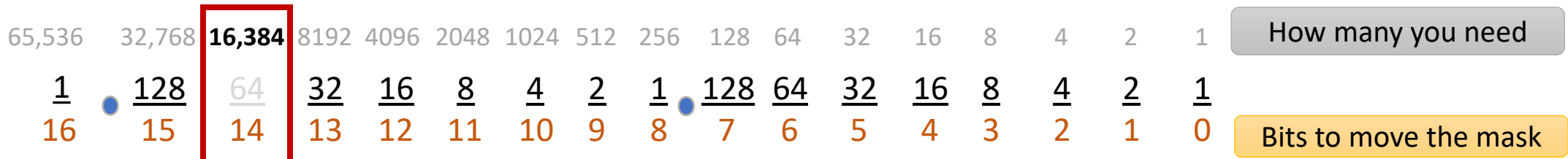
10.255.252.0

11.0.0.0



Start – 10.0.0.0 / 8





Start – 10.0.0.0 / 8

Need 10,000 subnets

What is the new subnet mask? /22 (255.255.252.0)

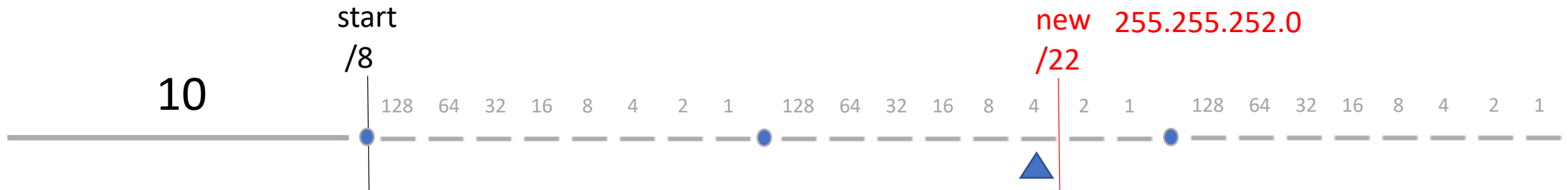
What is the delta, and in which octet? 4, 3rd

What is the subnet ID for the first two subnets?

What are the next two subnets after 10.0.40.0?

What are the next two subnets after 10.40.0.0?

What is the last subnet ID?



10.0.0.0	10.0.4.0
10.0.0.1	10.0.4.1
10.0.3.254	10.0.7.254
10.0.3.255	10.0.7.255

10.0.44.0	10.0.48.0
10.0.44.1	10.0.48.1
10.0.47.254	10.0.51.254
10.0.47.255	10.0.51.255

10.40.4.0	10.40.8.0
10.40.4.1	10.40.8.1
10.40.7.254	10.40.11.254
10.40.7.255	10.40.11.255

10.255.252.0
10.255.252.1
10.255.255.254
10.255.255.255

